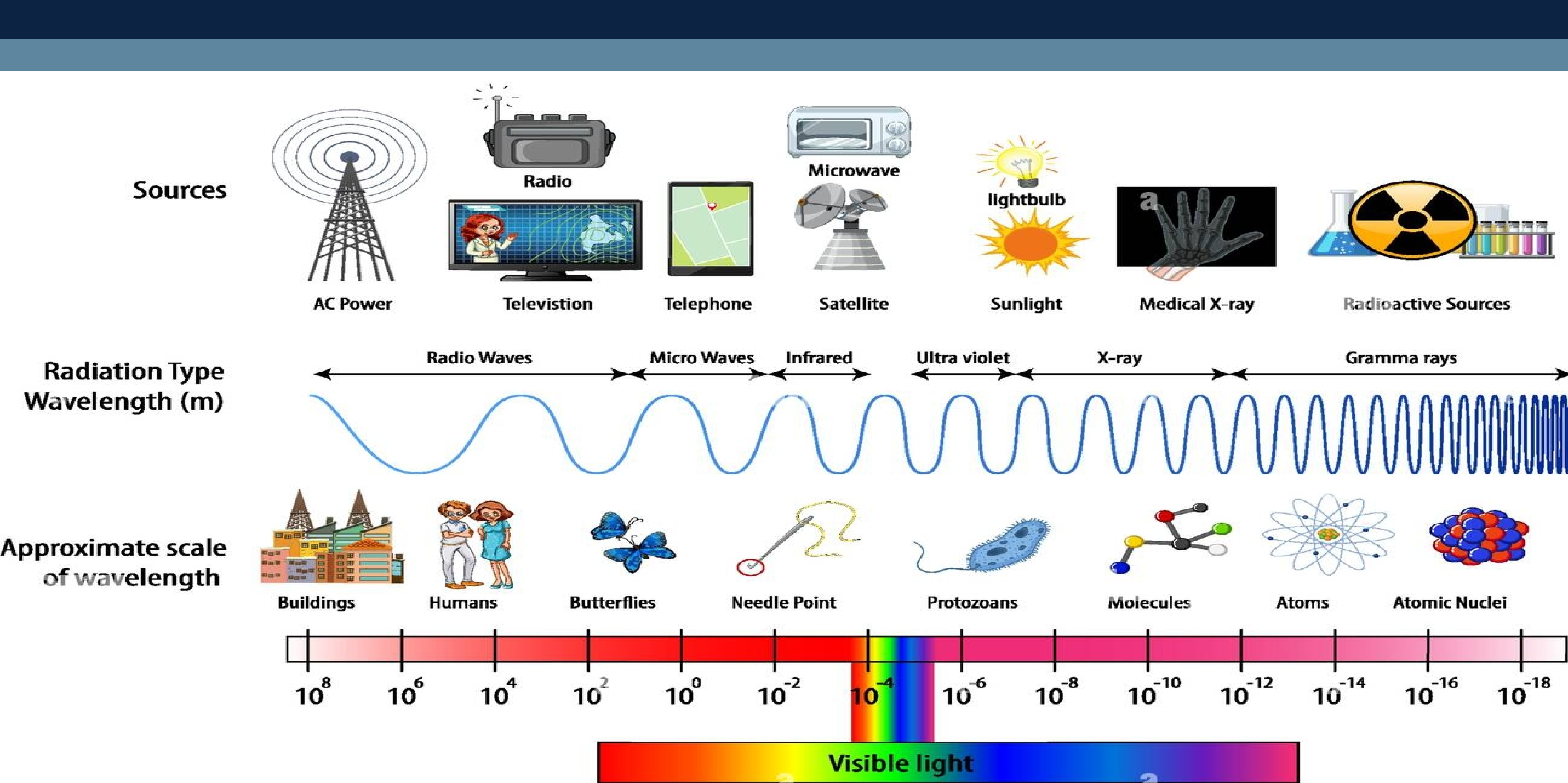
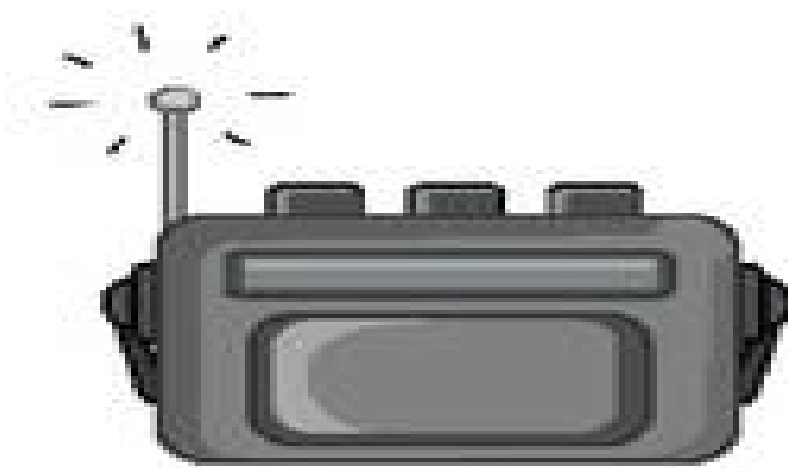






Radio



Microwave



ELECTROMAGNETIC WAVES

CHAPTER EIGHT CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

Approximate scale
of wavelength



Buildings



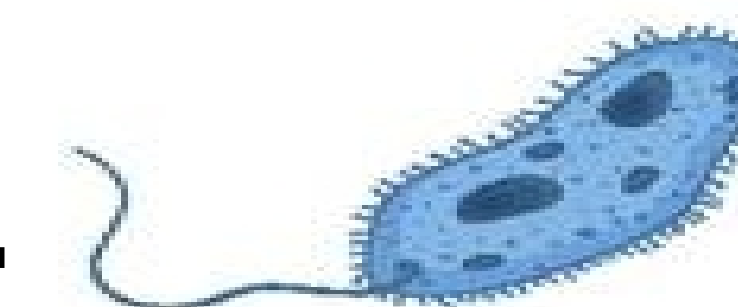
Humans



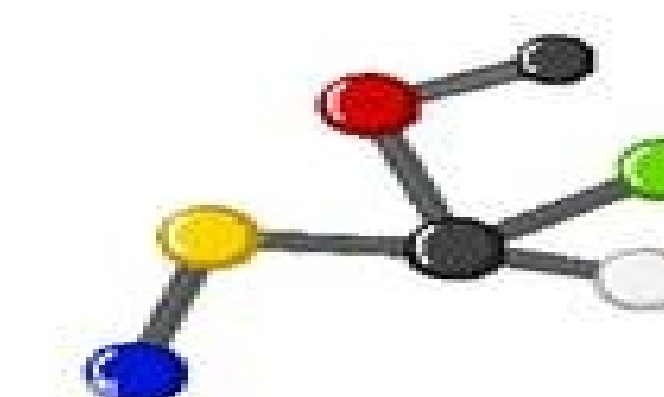
Butterflies



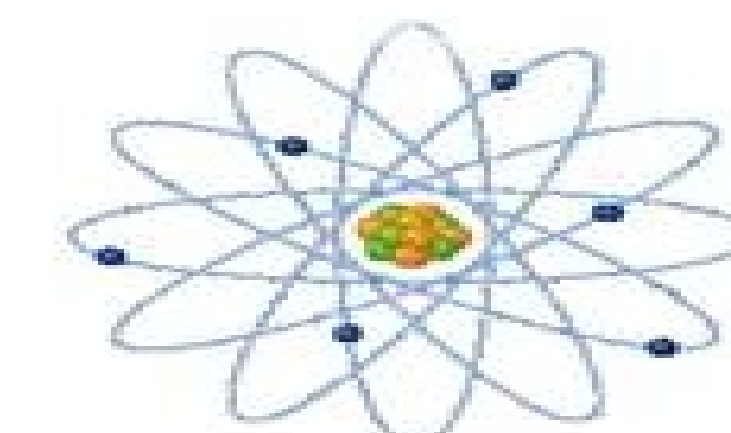
Needle Point



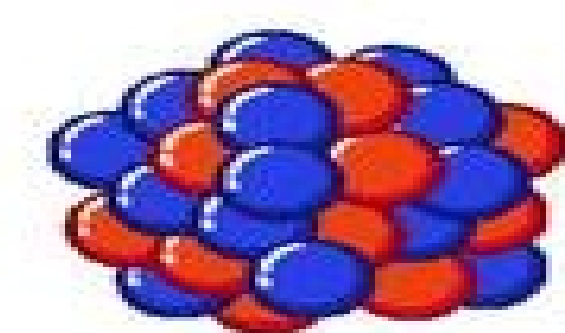
Protozoans



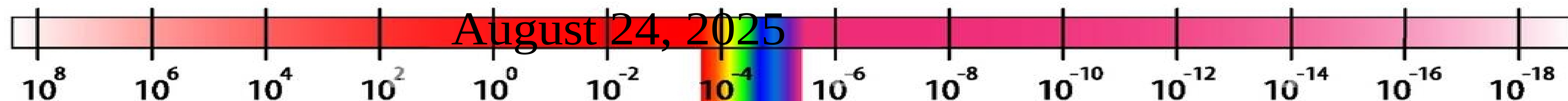
Molecules



Atoms



Atomic Nuclei



August 24, 2025

Visible light

Learning Objectives:

- Introduction
- Basic idea of displacement current,
- Electromagnetic waves, their characteristics, their transverse nature (qualitative idea only).
- Electromagnetic spectrum (radio waves, microwaves, infra-red, visible, ultraviolet, X-rays, gamma rays) including elementary facts about their uses.

INTRODUCTION:

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- It means that a **change in one field with time** (either electric or magnetic) **produces another field**.
- This idea led Maxwell to conclude that the **variation in electric and magnetic fields perpendicular to each other, produces electromagnetic disturbances** in space.

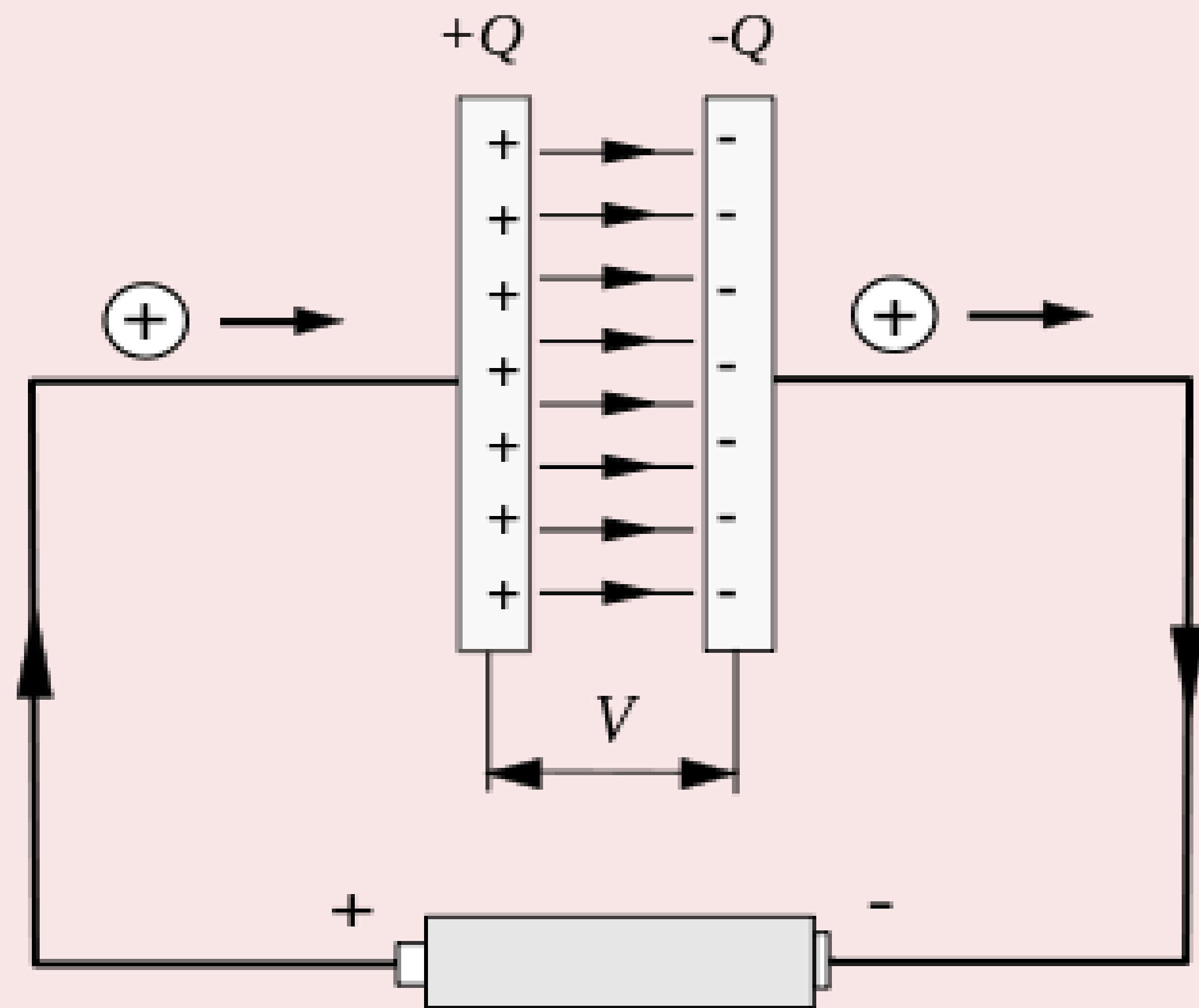
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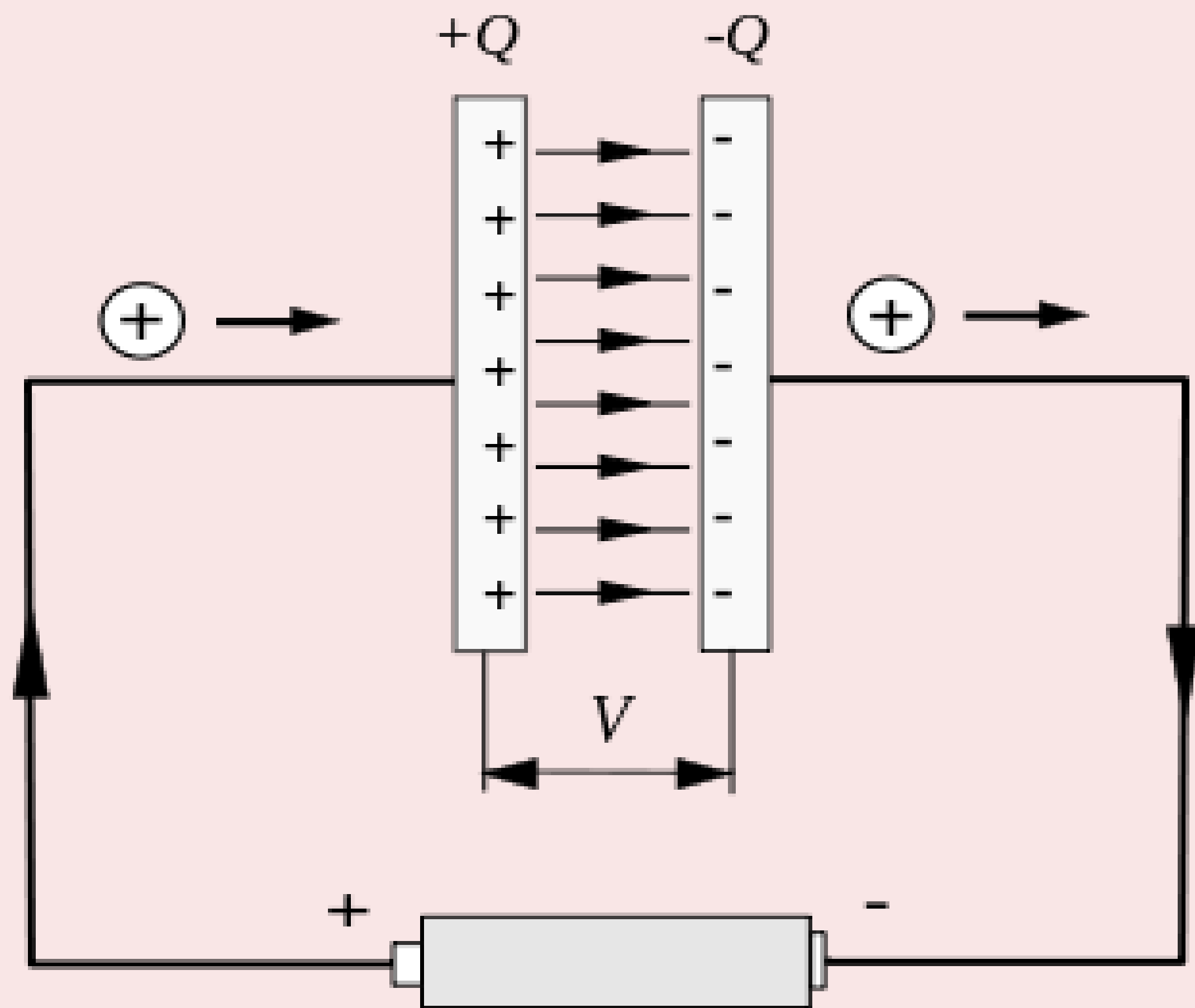
- **These disturbances** have the properties of a wave and **propagate through space without any material medium.**
- These waves are called **electromagnetic waves**. They travel in vacuum or free space with a velocity $3 \times 10^8 \text{ ms}^{-1}$

DISPLACEMENT CURRENT:



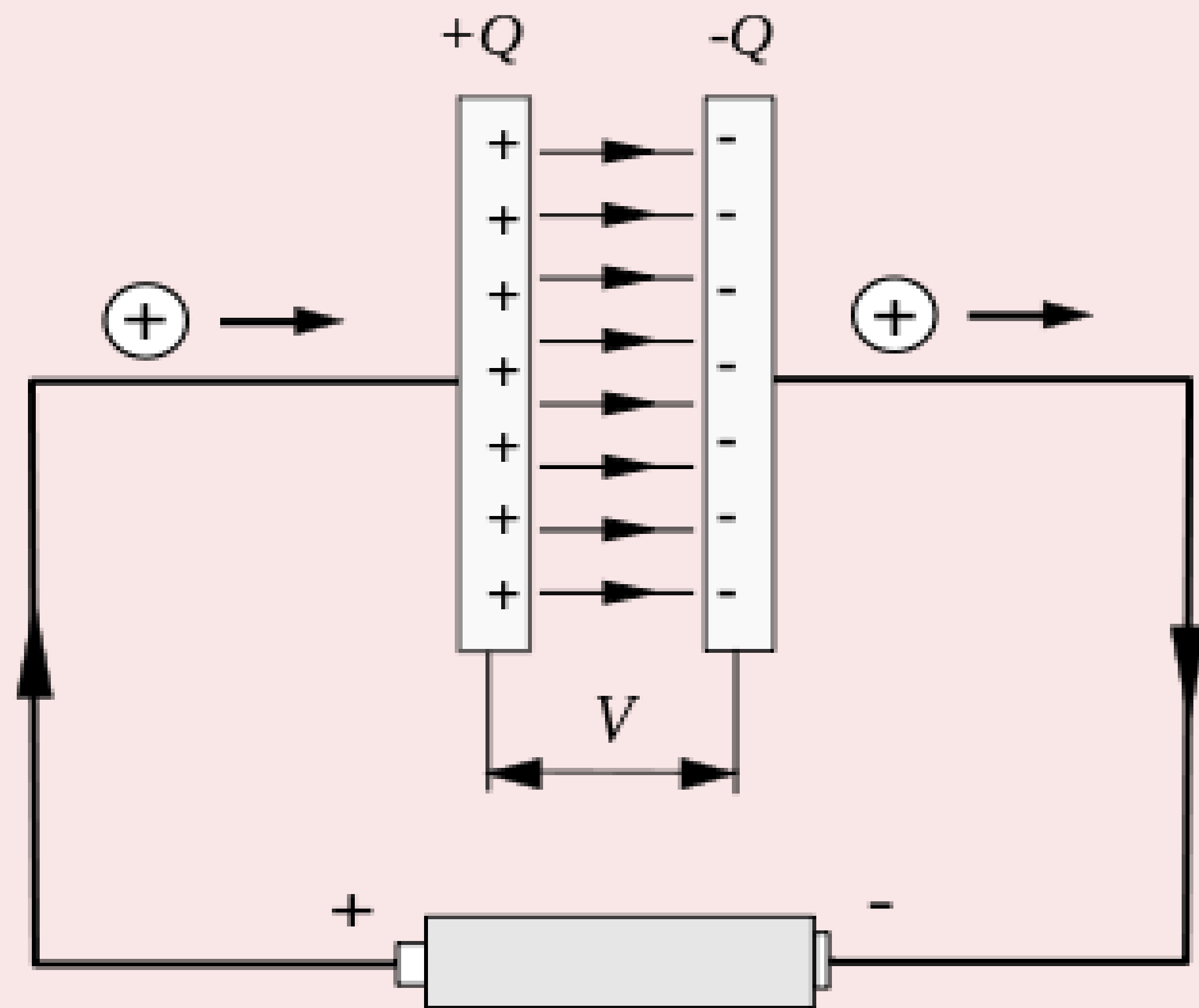
- We know that an electric current produces a magnetic field around it. **Maxwell showed that a changing electric field must also produce a magnetic field .**

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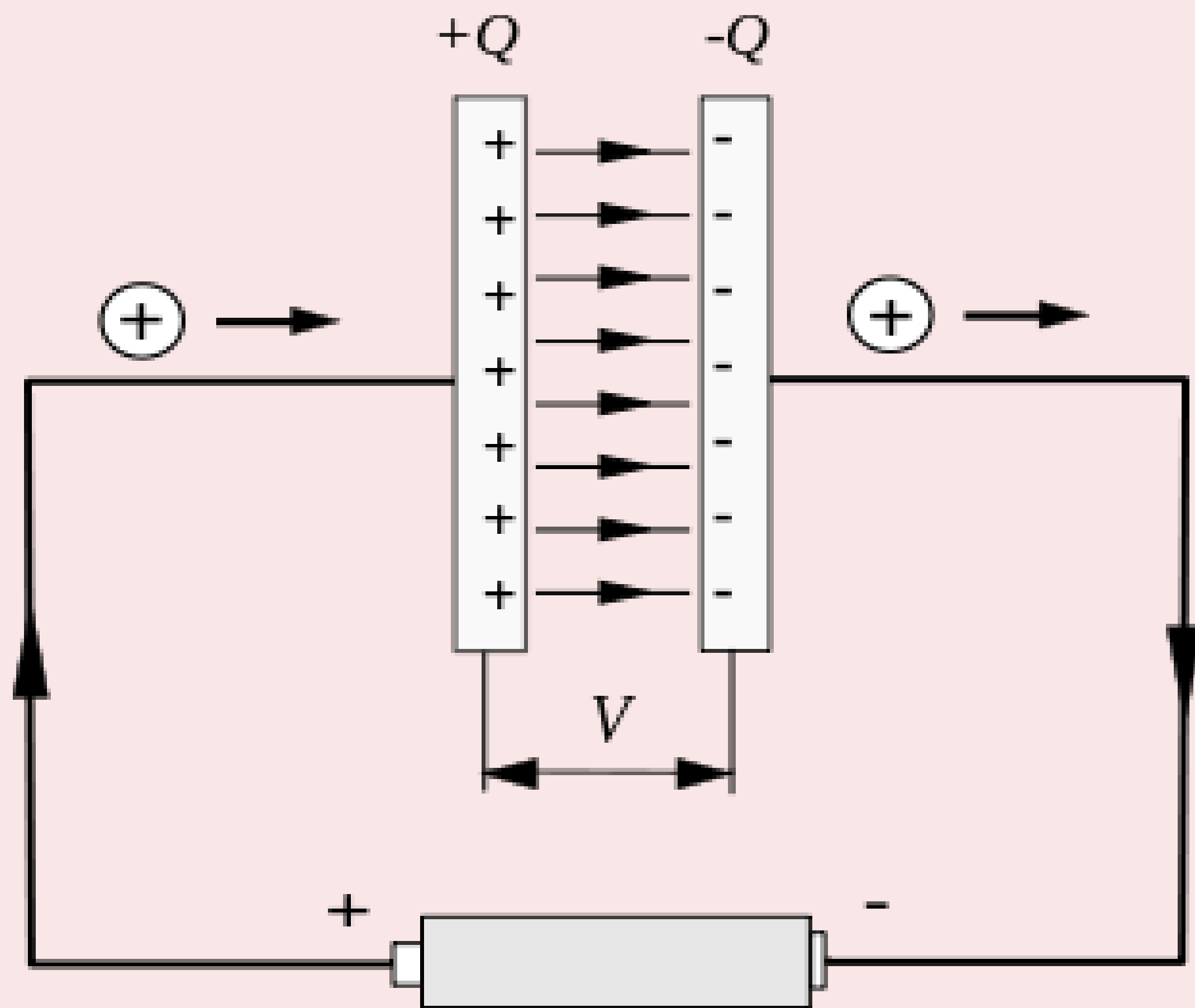
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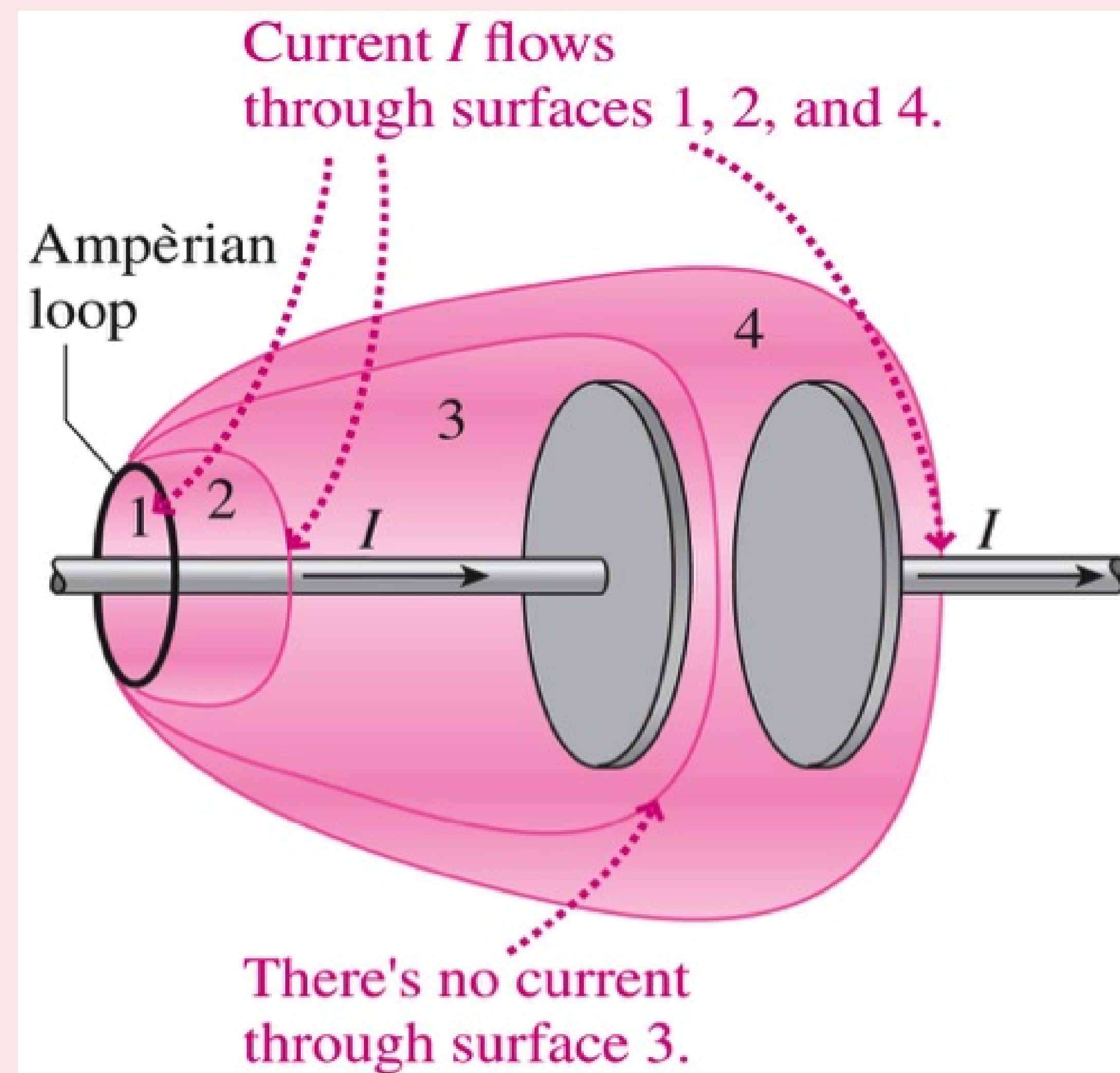
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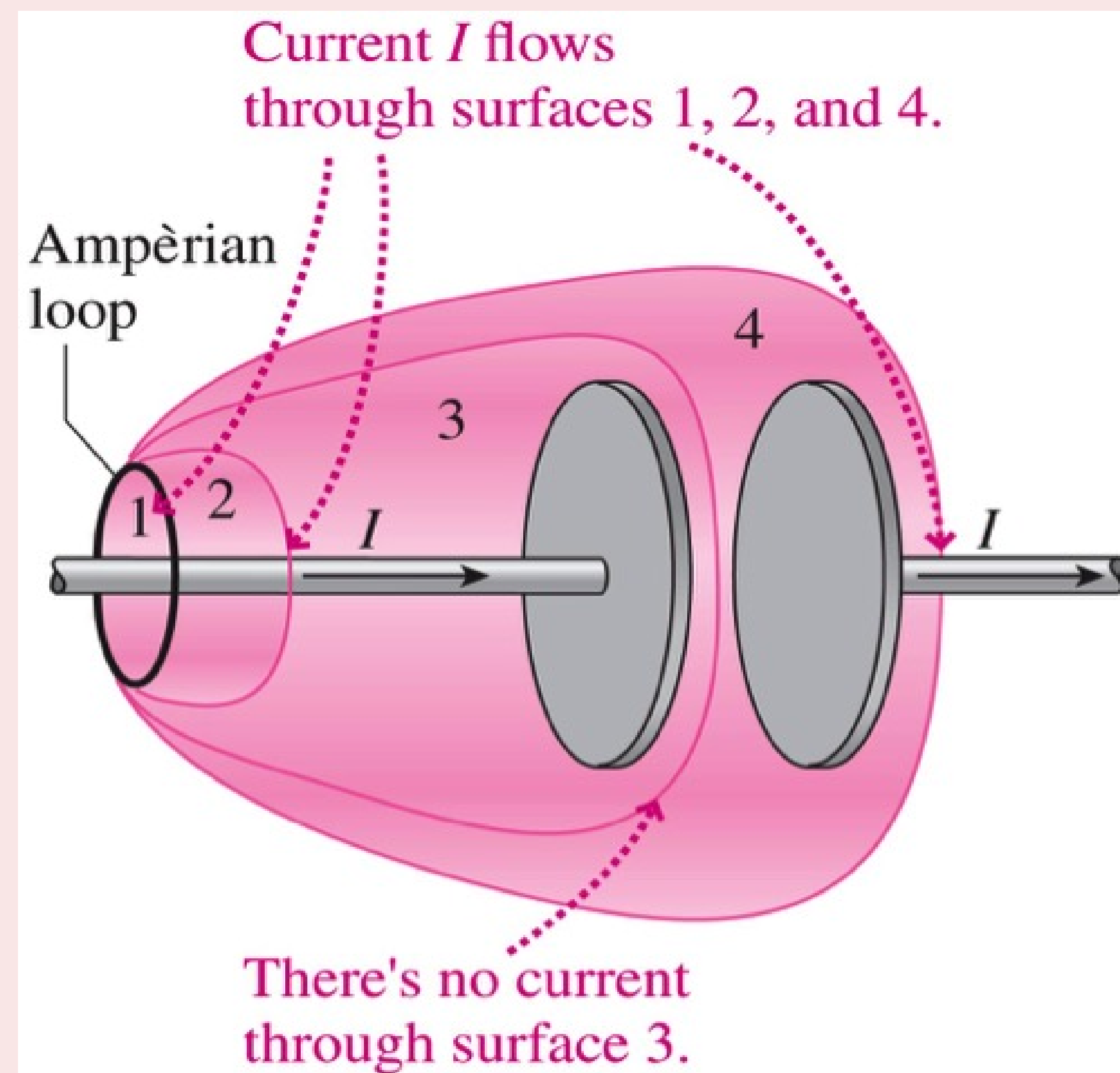
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \dots (1)$$

DISPLACEMENT CURRENT:



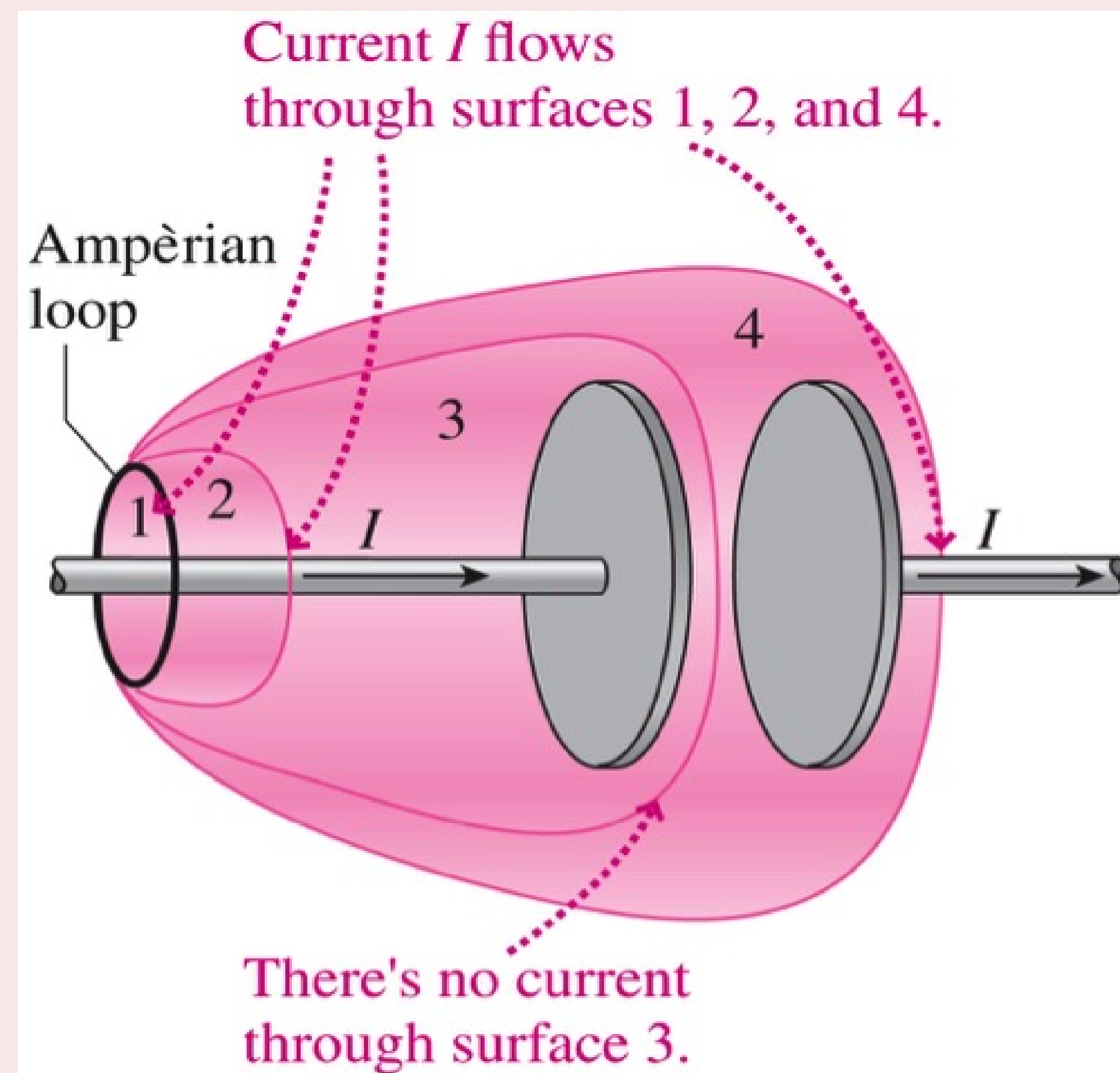
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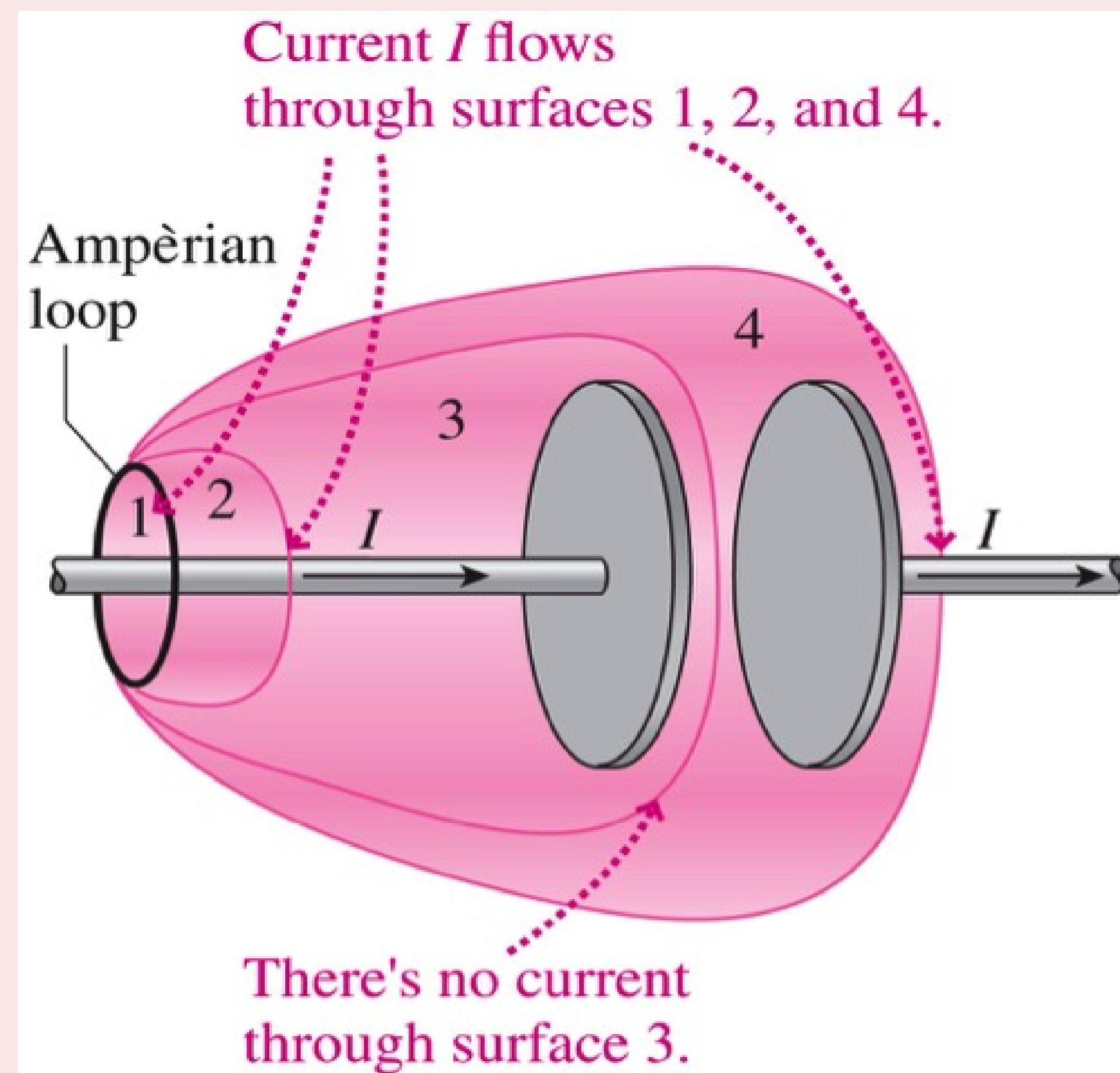
- Consider a parallel plate **capacitor C** which is a part of circuit through which a time-dependent **current I** flows.
- To find the **magnetic field at a point P**, in a region outside the parallel plate capacitor, let us **consider a plane circular loop of radius r** whose plane is perpendicular to the direction of the current-carrying wire, and which is **centred symmetrically** with respect to the wire.

DISPLACEMENT CURRENT:



- By symmetry, the **magnetic field is directed along the circumference** of the circular loop and is the **same in magnitude at all points on the loop**. If B is the magnitude of the field, then

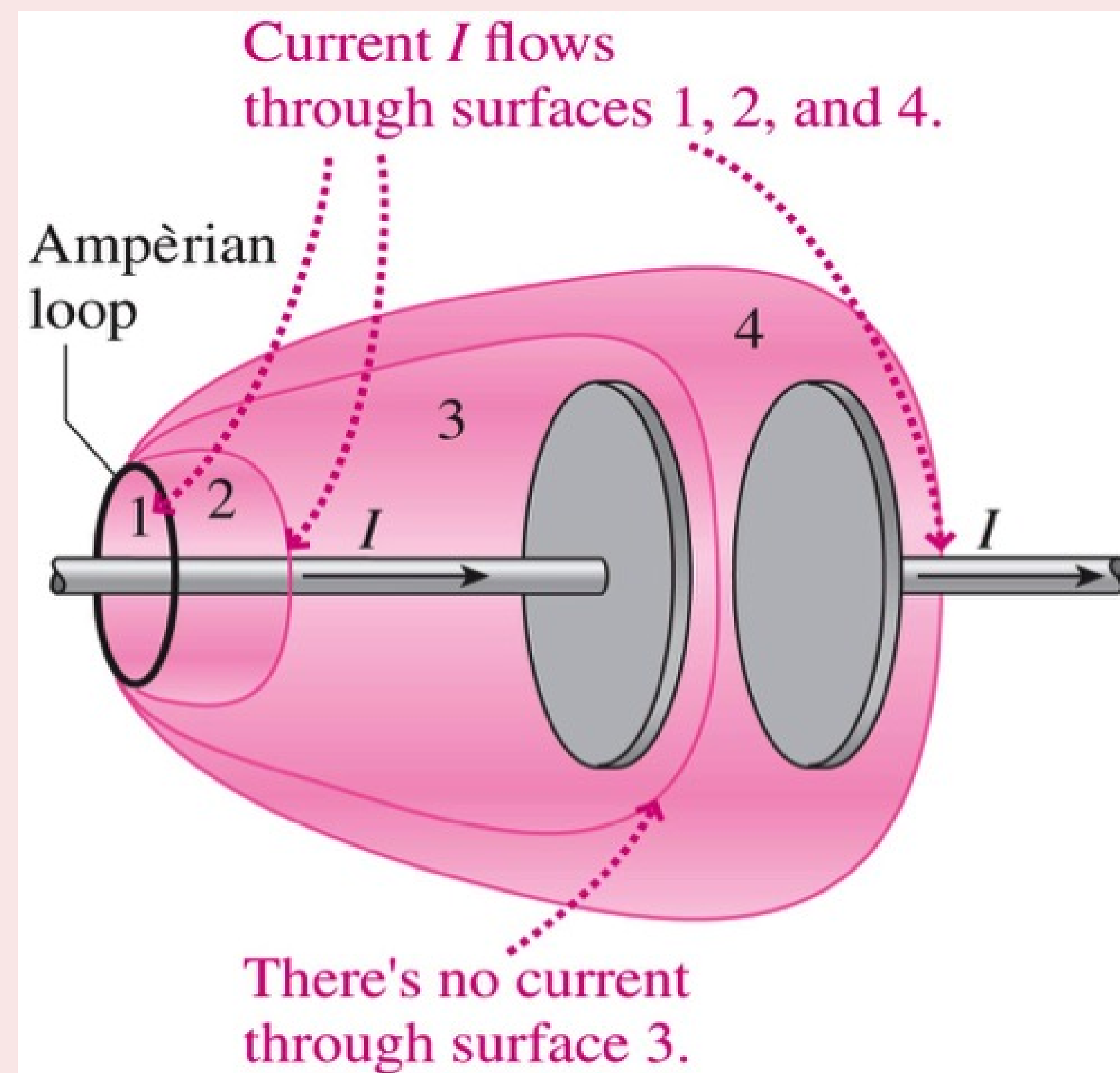
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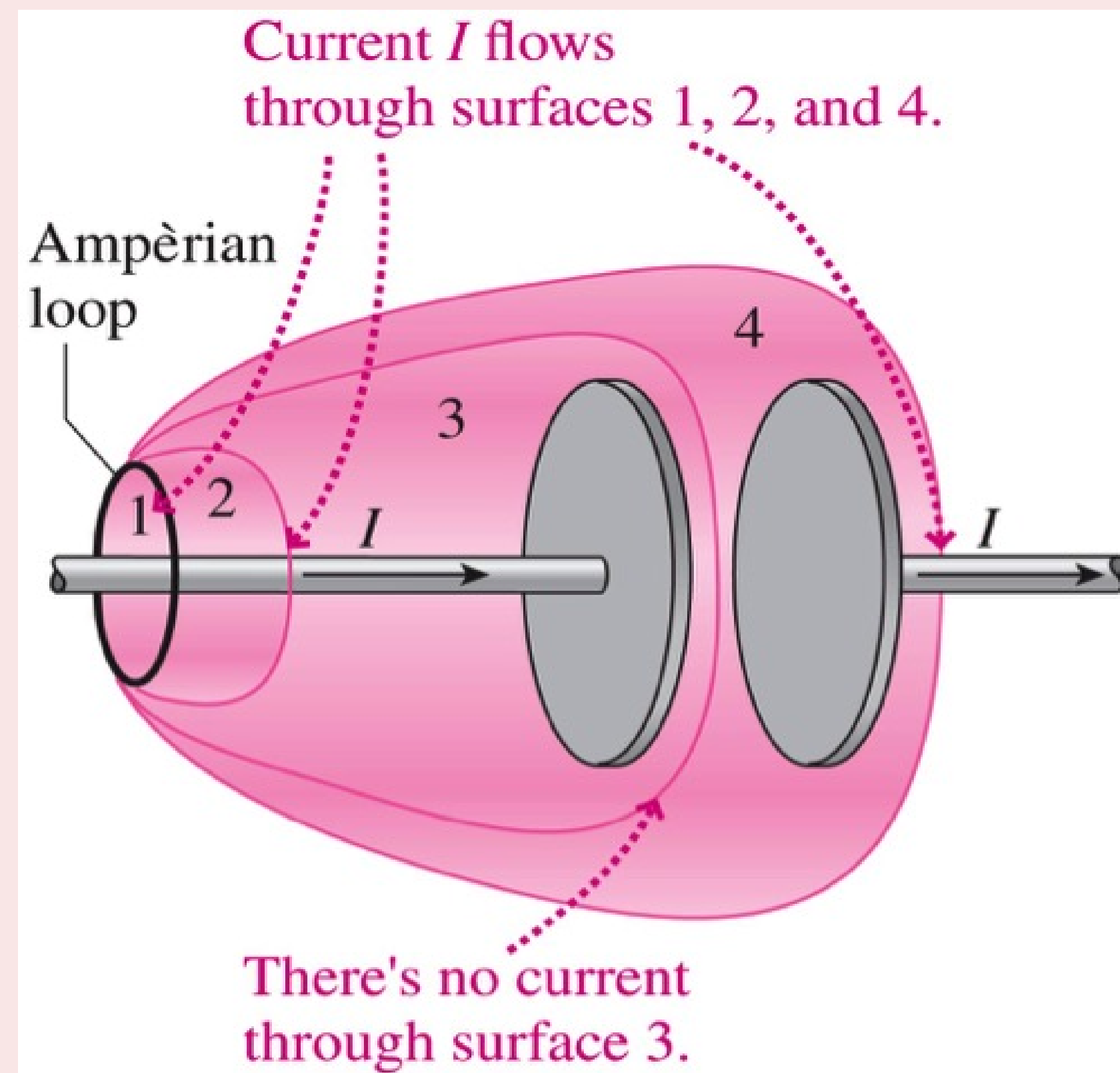
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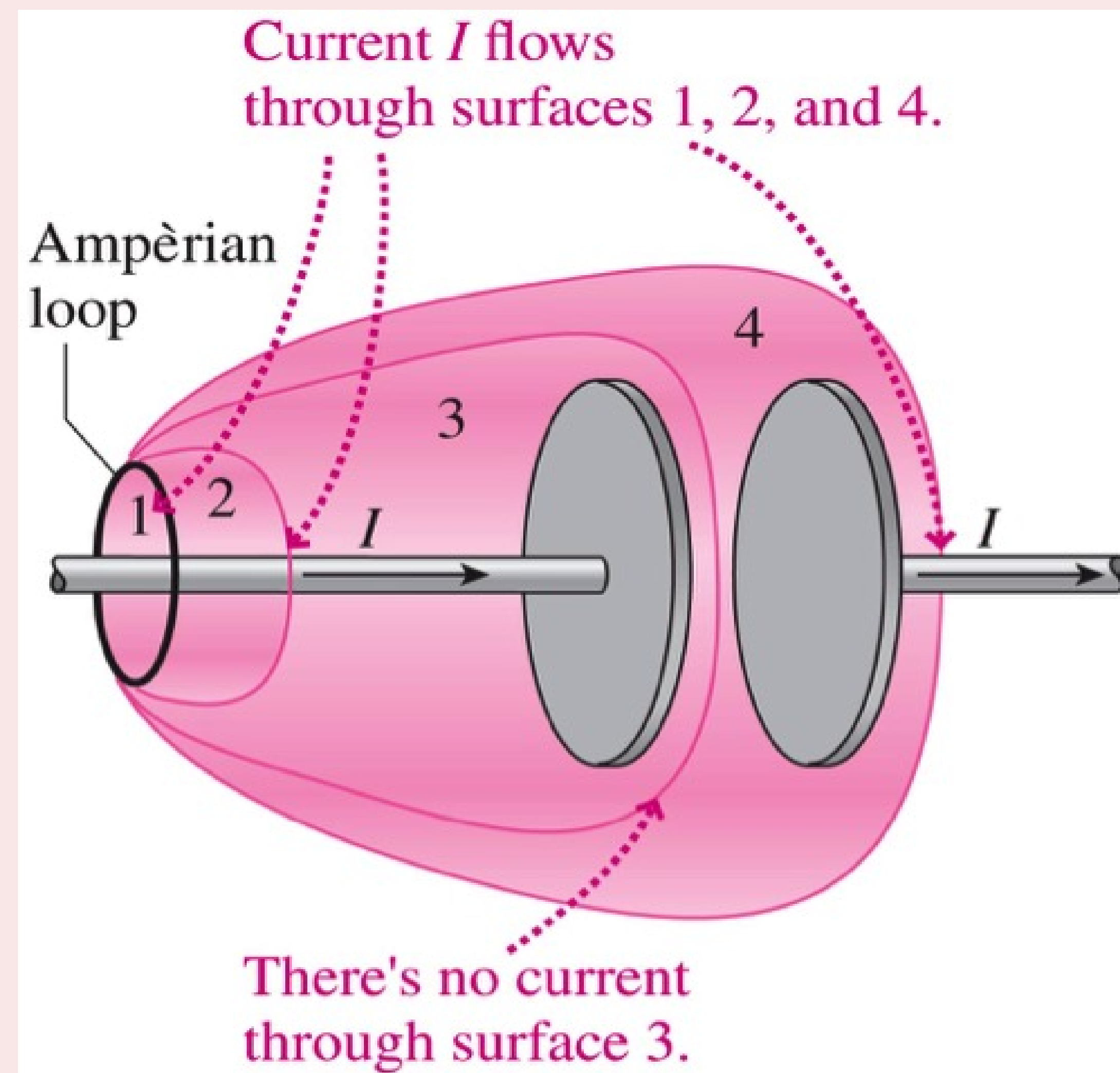
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- There are **infinitely many surfaces that can be attached to any loop**, and **Ampere's law** stated in Equation (1) is **independent of the choice of surface**.

DISPLACEMENT CURRENT:



- **Surface 1, 2, and 4 gives a non-zero value for the enclosed current I , whereas surface 3 gives zero for the enclosed current because no current passes through it:**

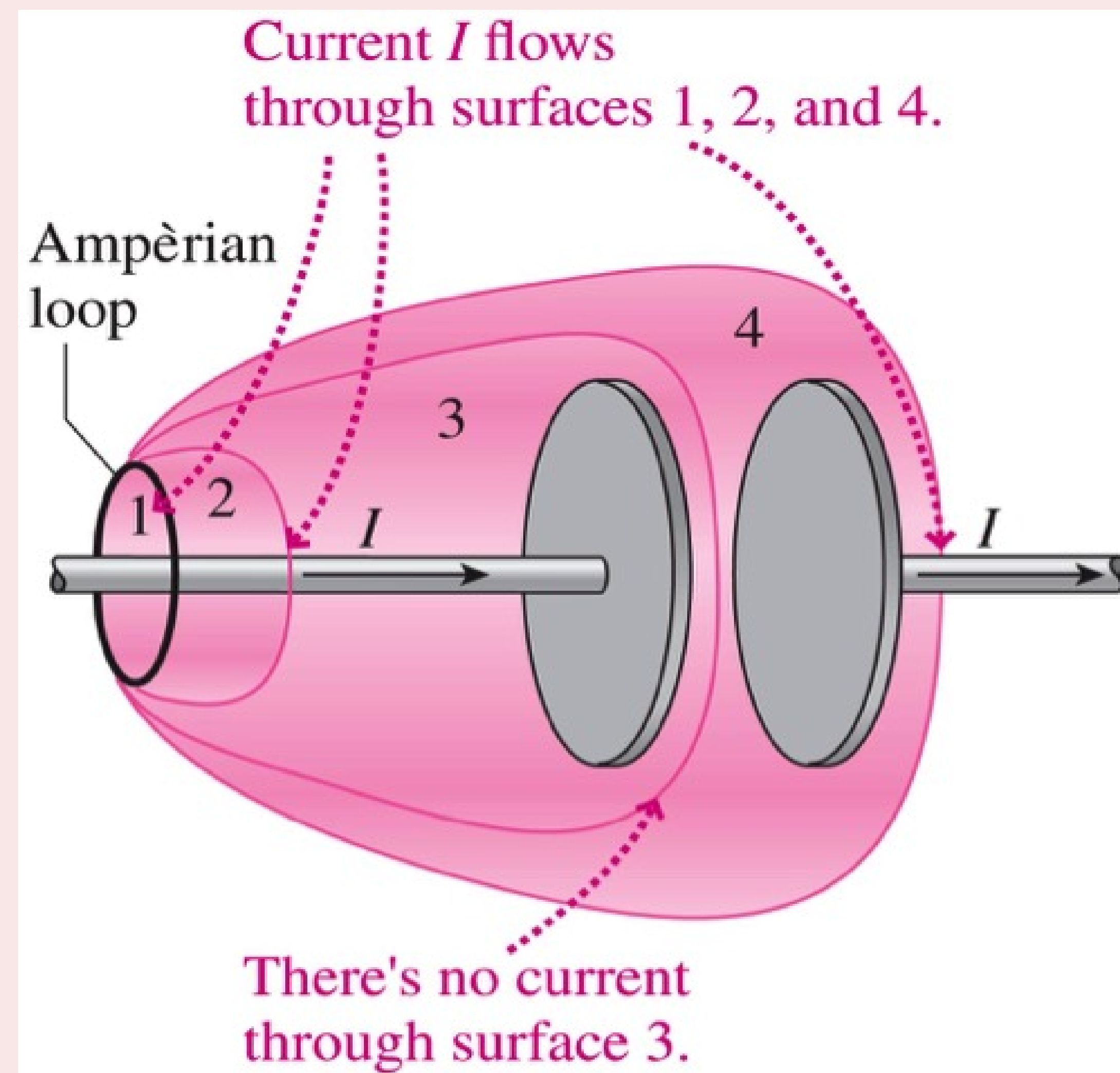
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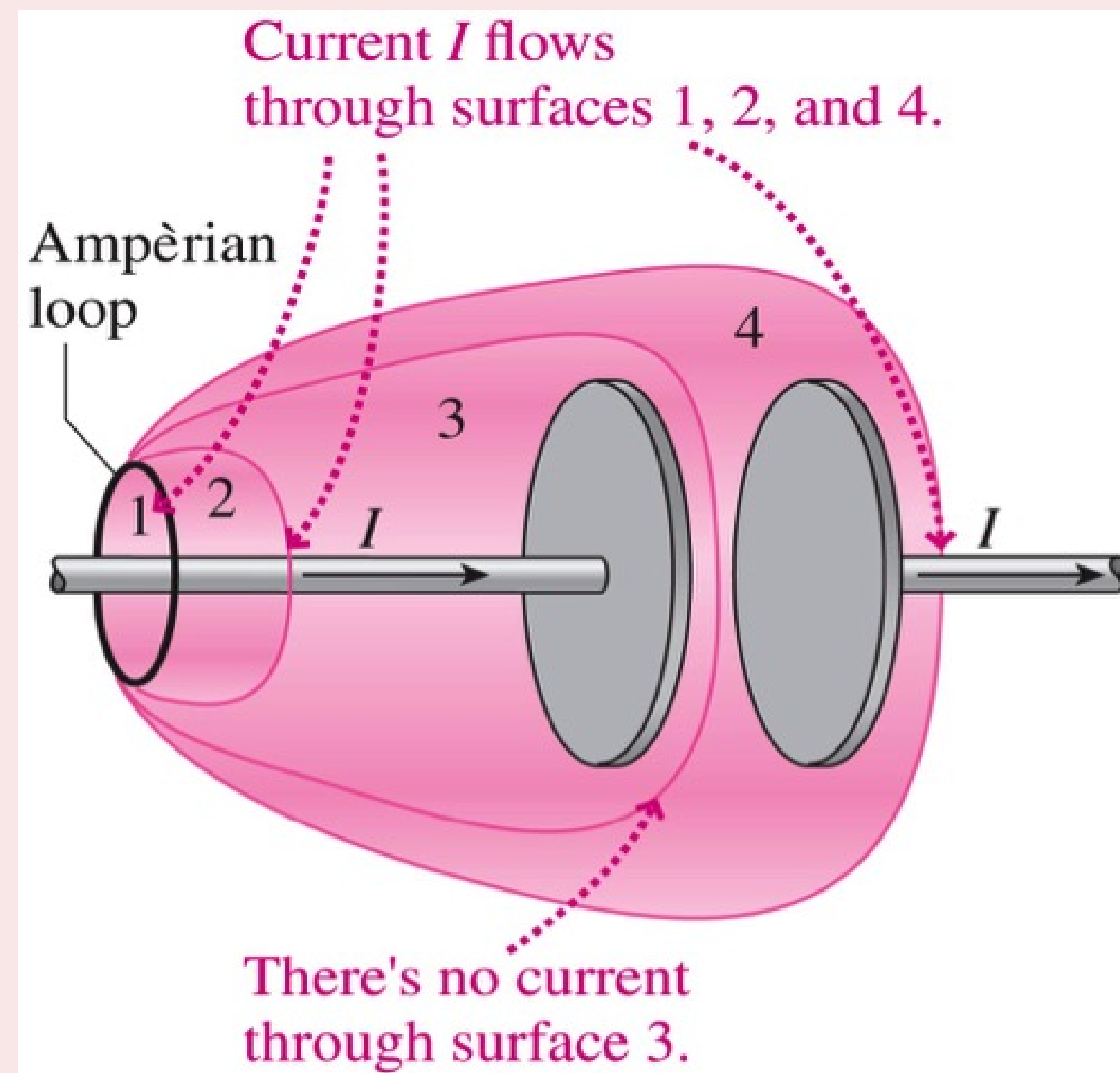


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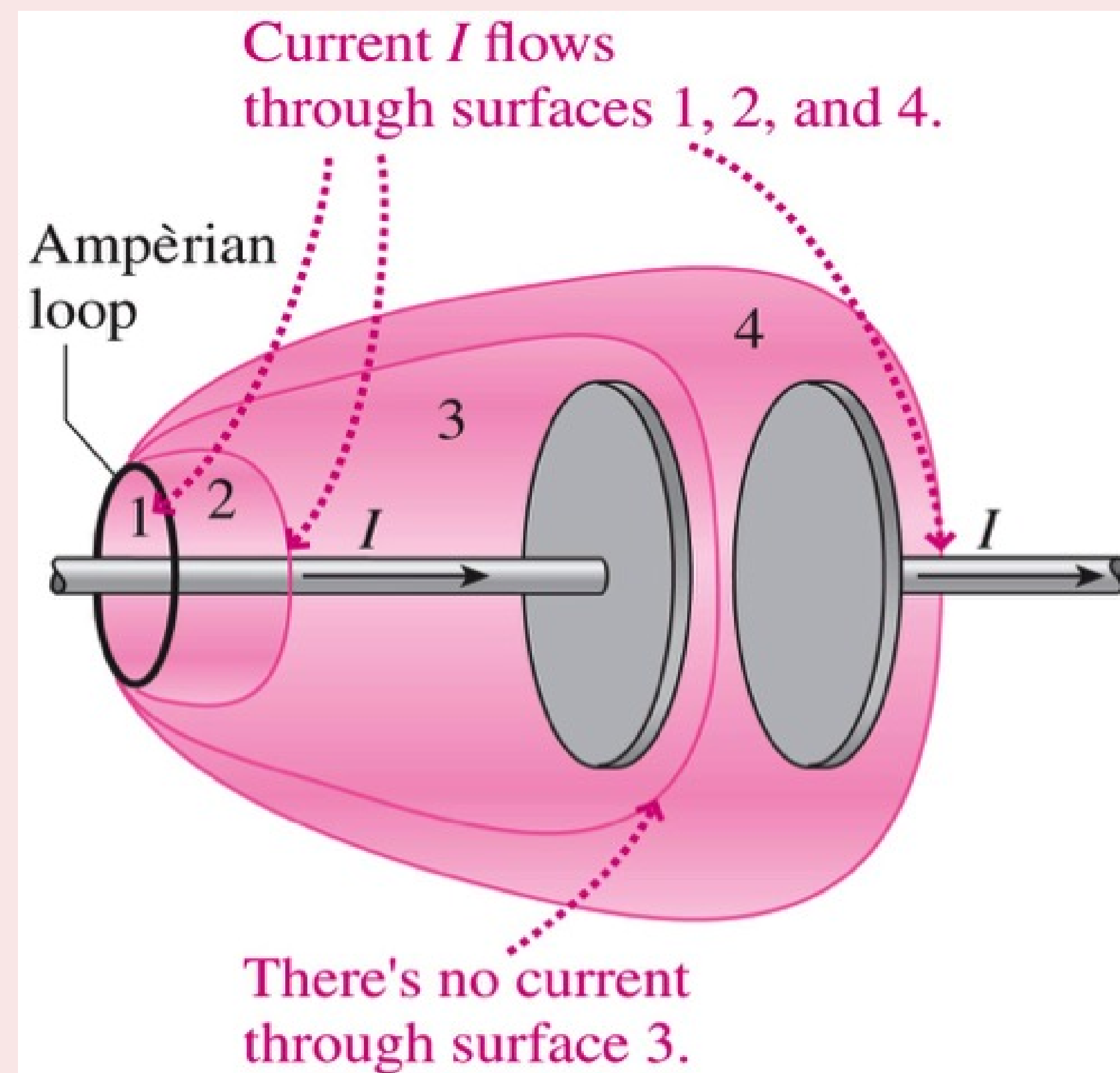
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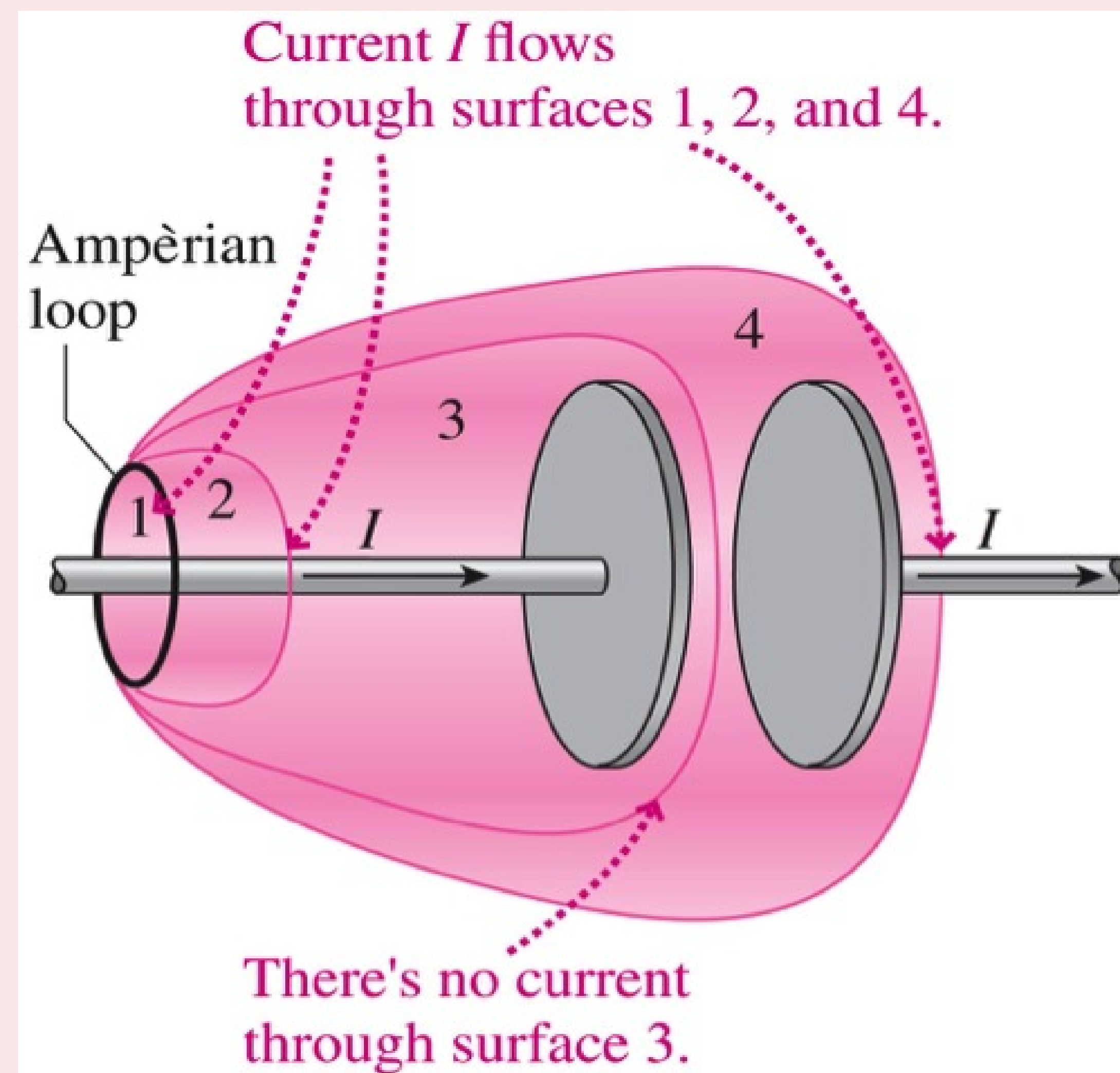
- **On applying Ampere's circuital law to such surfaces with the same perimeter, we find that the LHS term in eq (2) is the same, but the RHS term is zero, since no current passes through the surfaces.**

DISPLACEMENT CURRENT:



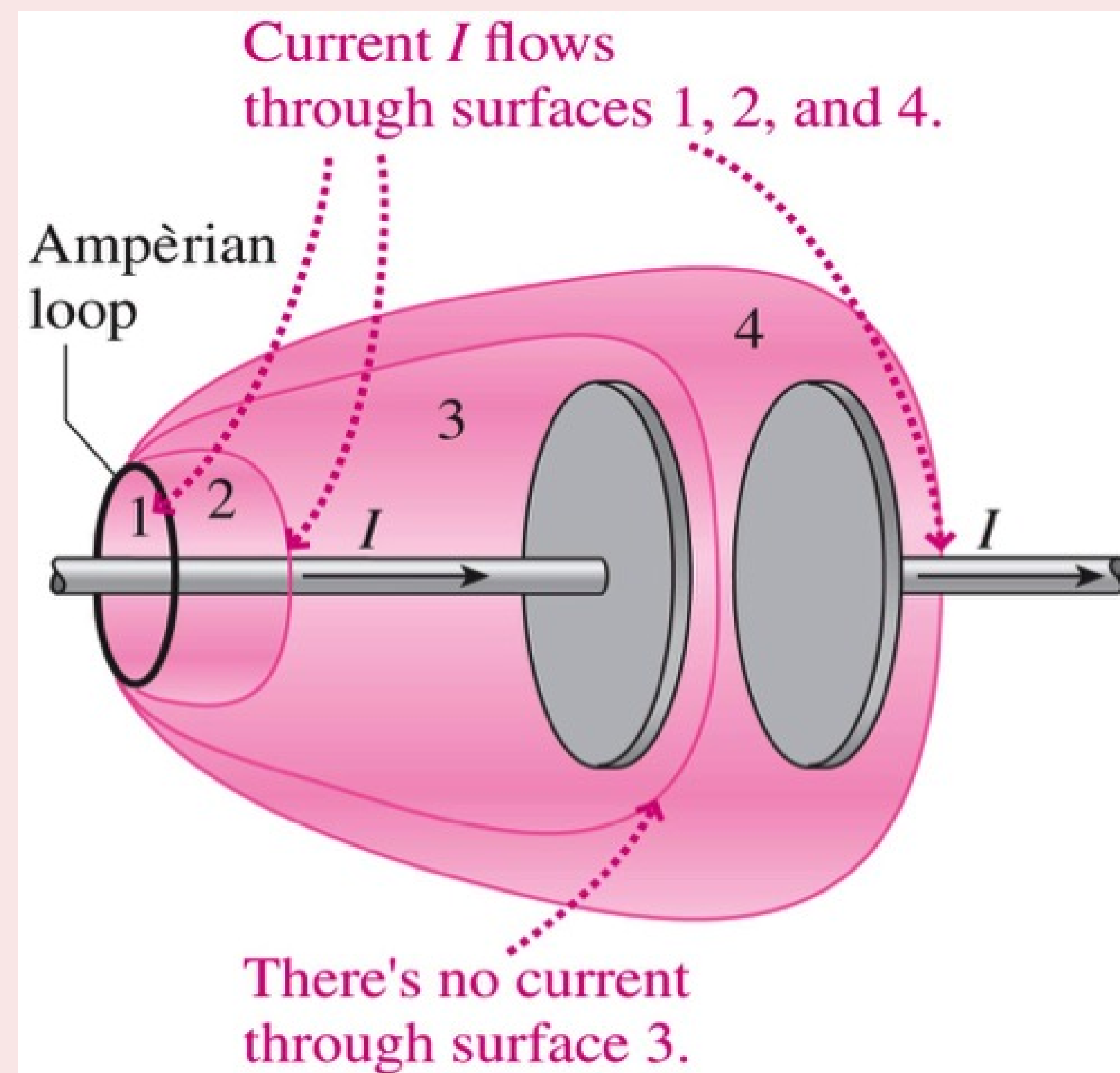
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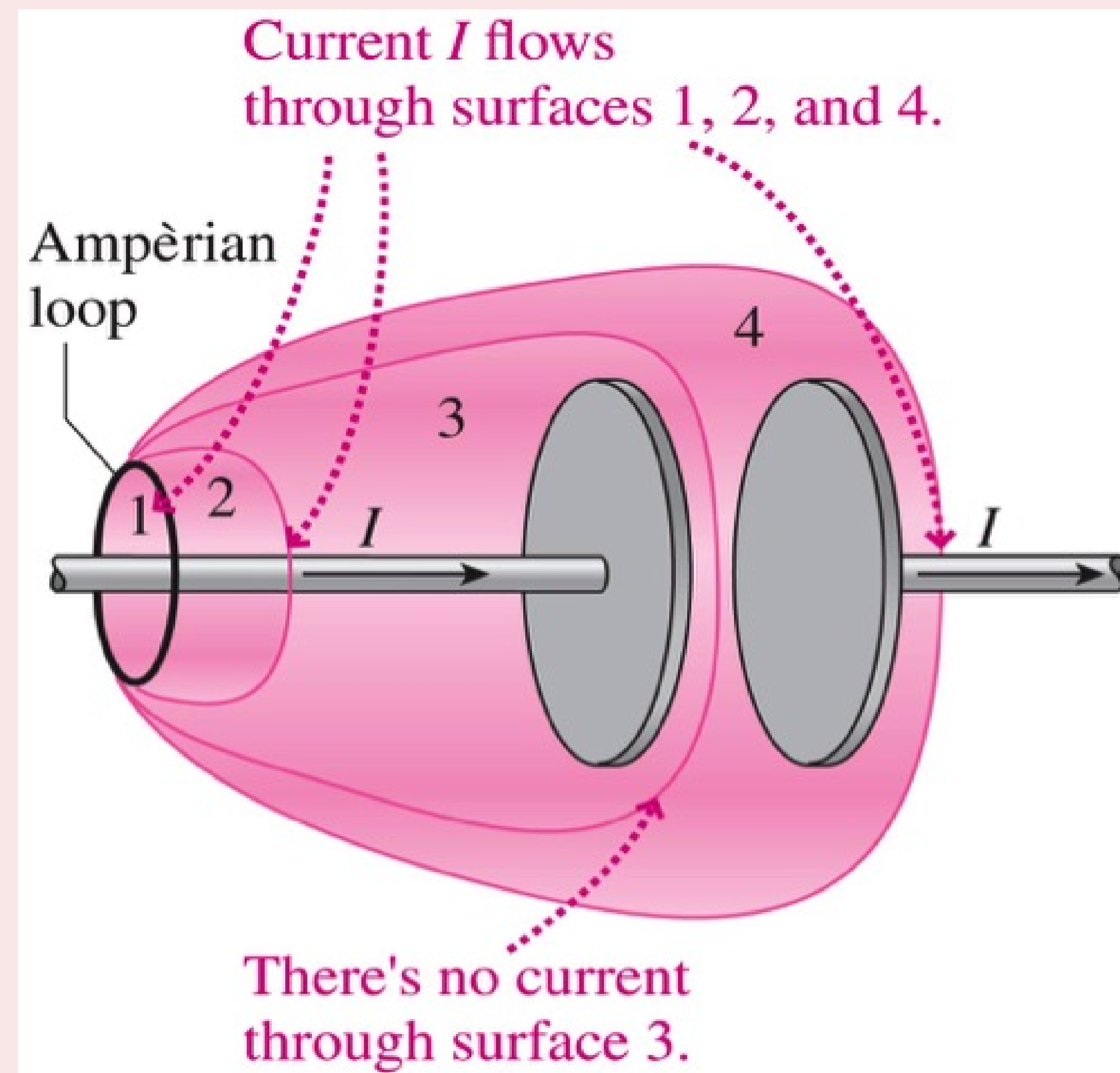
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- If $A \rightarrow$ area of the plates of the capacitor and $Q \rightarrow$ total charge on each plate of the capacitor

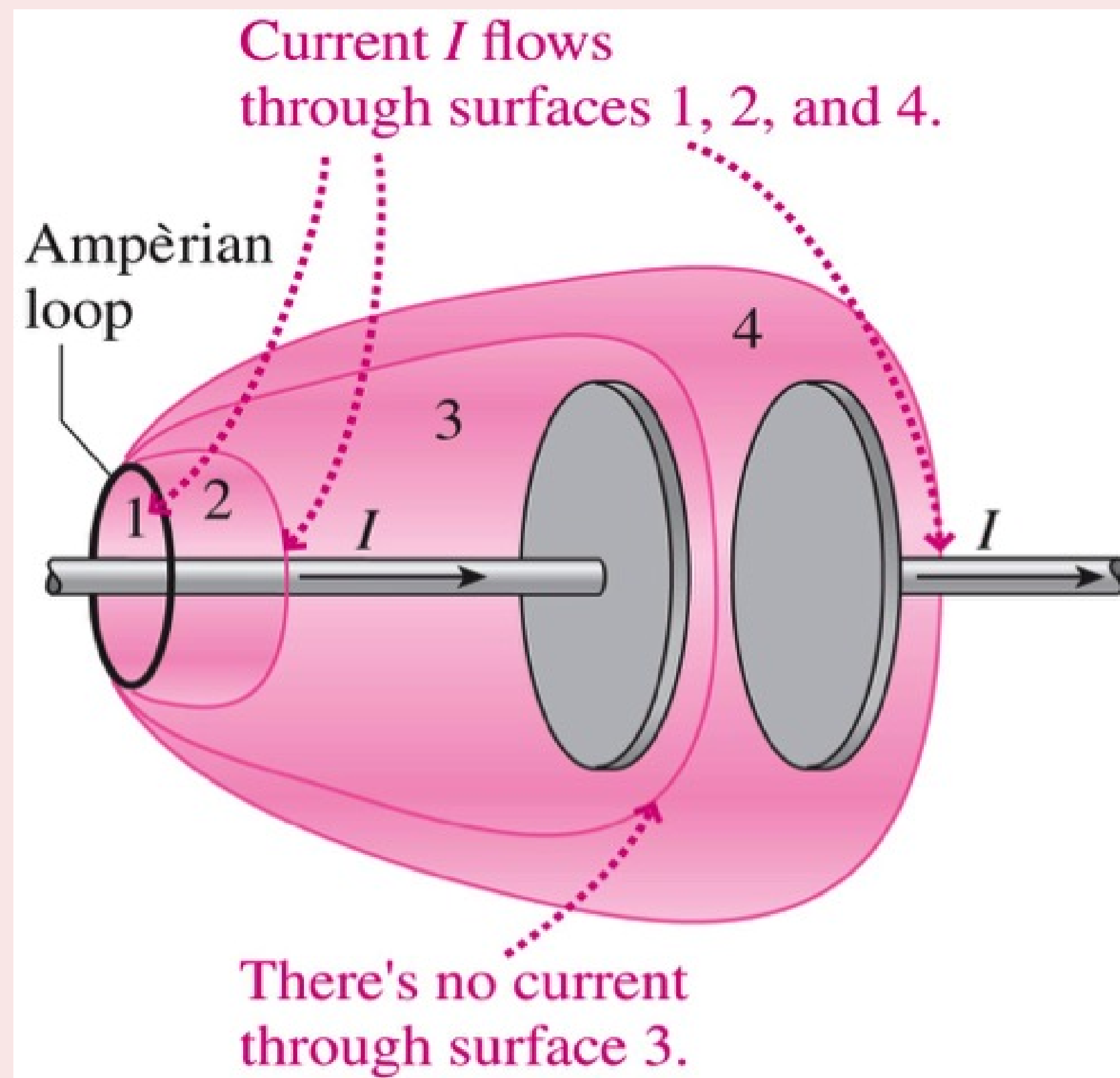
DISPLACEMENT CURRENT:



- The electric field E between the plates is given by

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0} \dots (3)$$

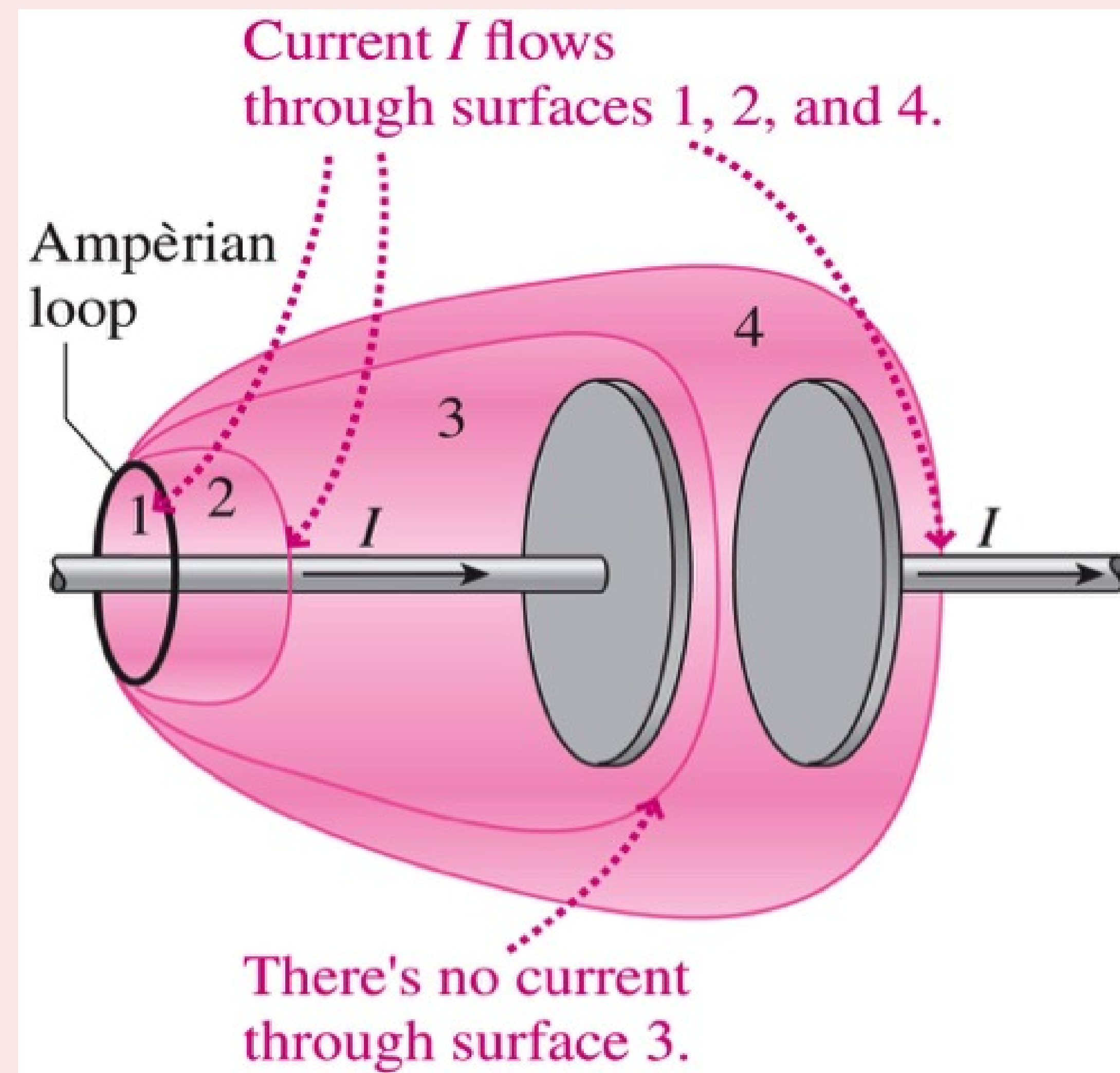
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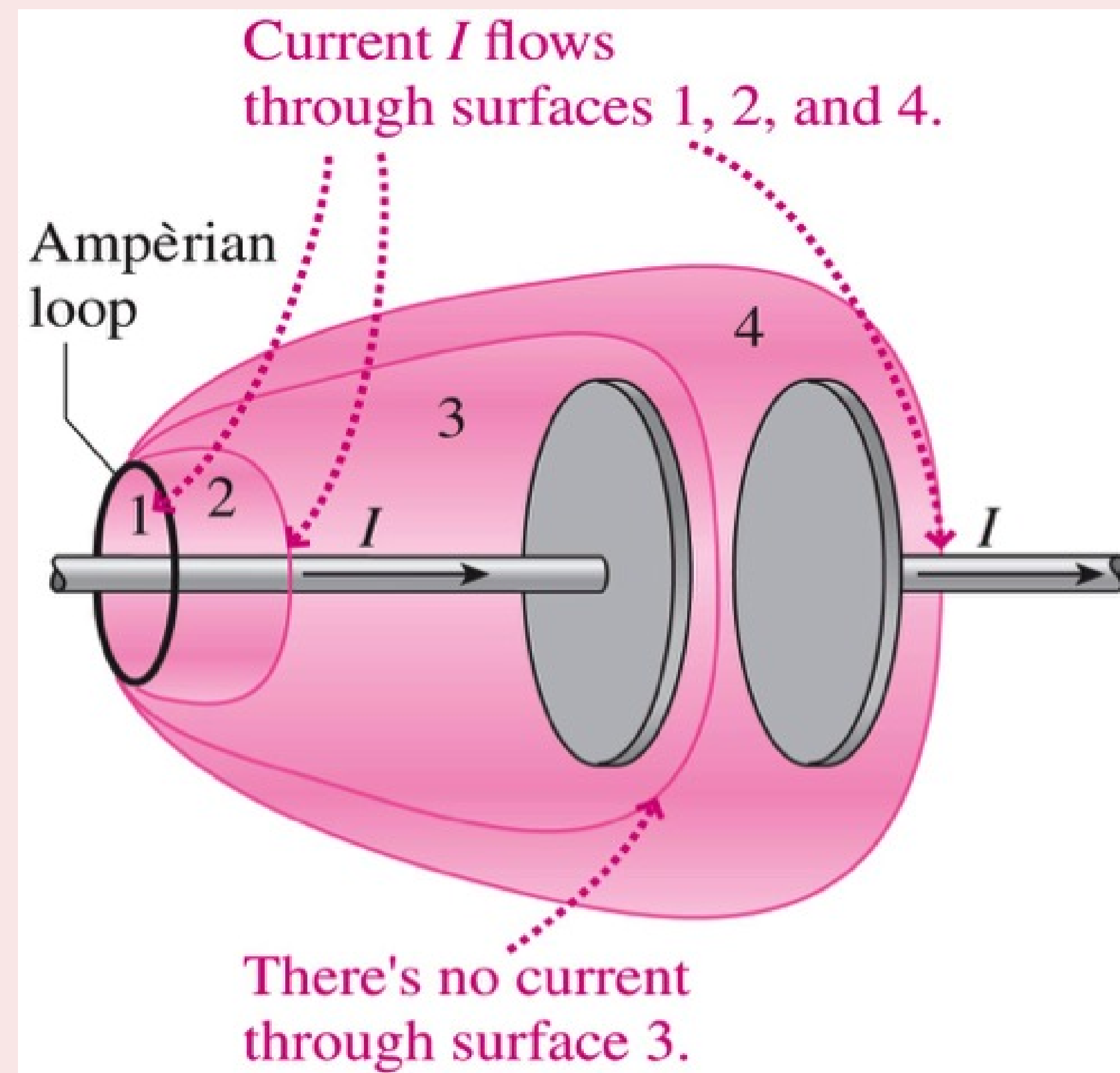
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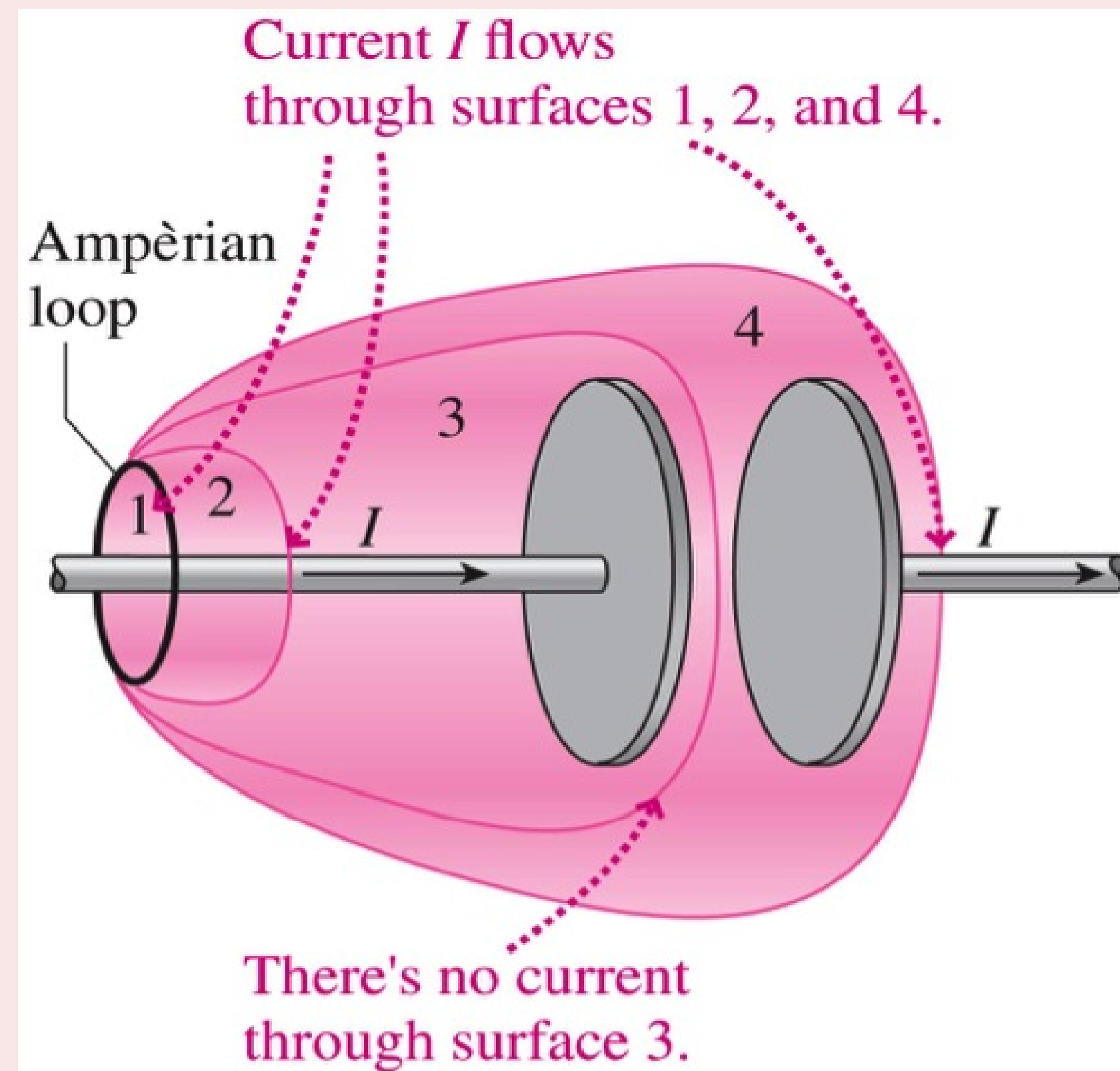
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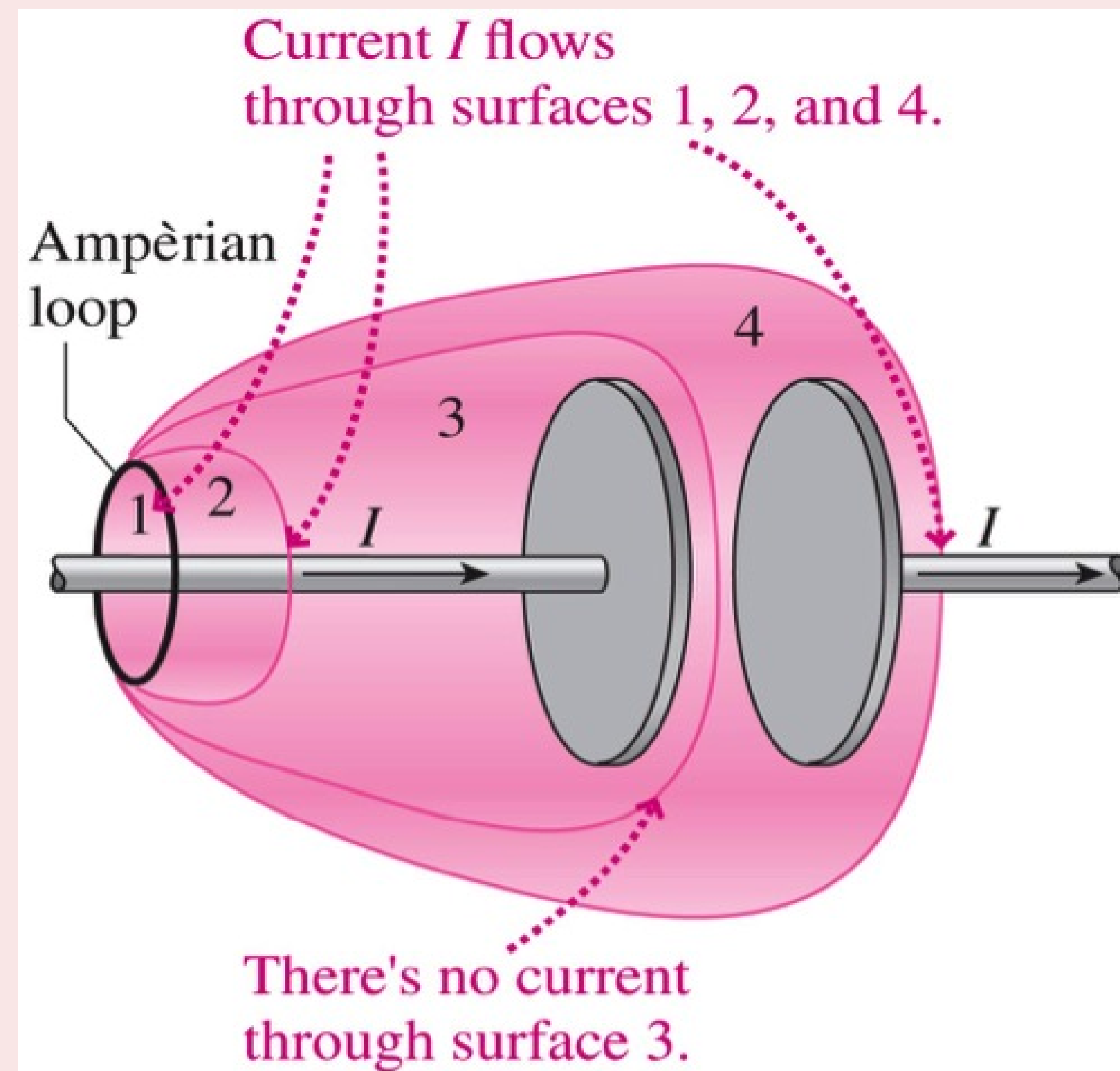
$$\Phi_E = EA = \frac{Q}{\epsilon_0}$$
$$\text{Or } Q = \epsilon_0 \Phi_E \dots (4)$$

DISPLACEMENT CURRENT:



- Since the **charge Q on the capacitor plates changes with time**, there is a **current I** which is given by

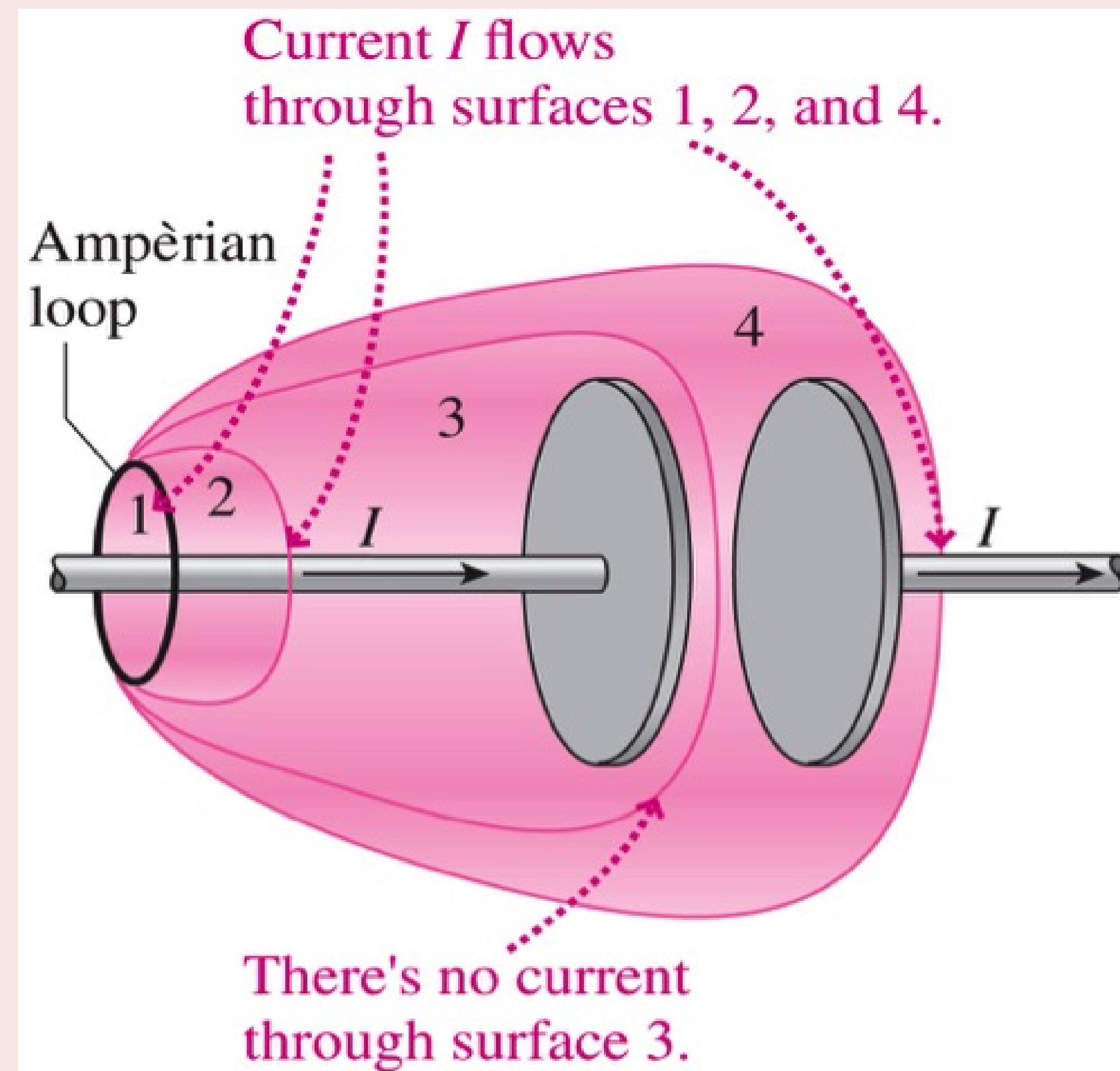
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- Since the **charge Q on the capacitor plates changes with time**, there is a **current I** which is given by

$$I = \frac{dQ}{dt} = \epsilon_0 \frac{d\Phi_E}{dt} \dots (5)$$

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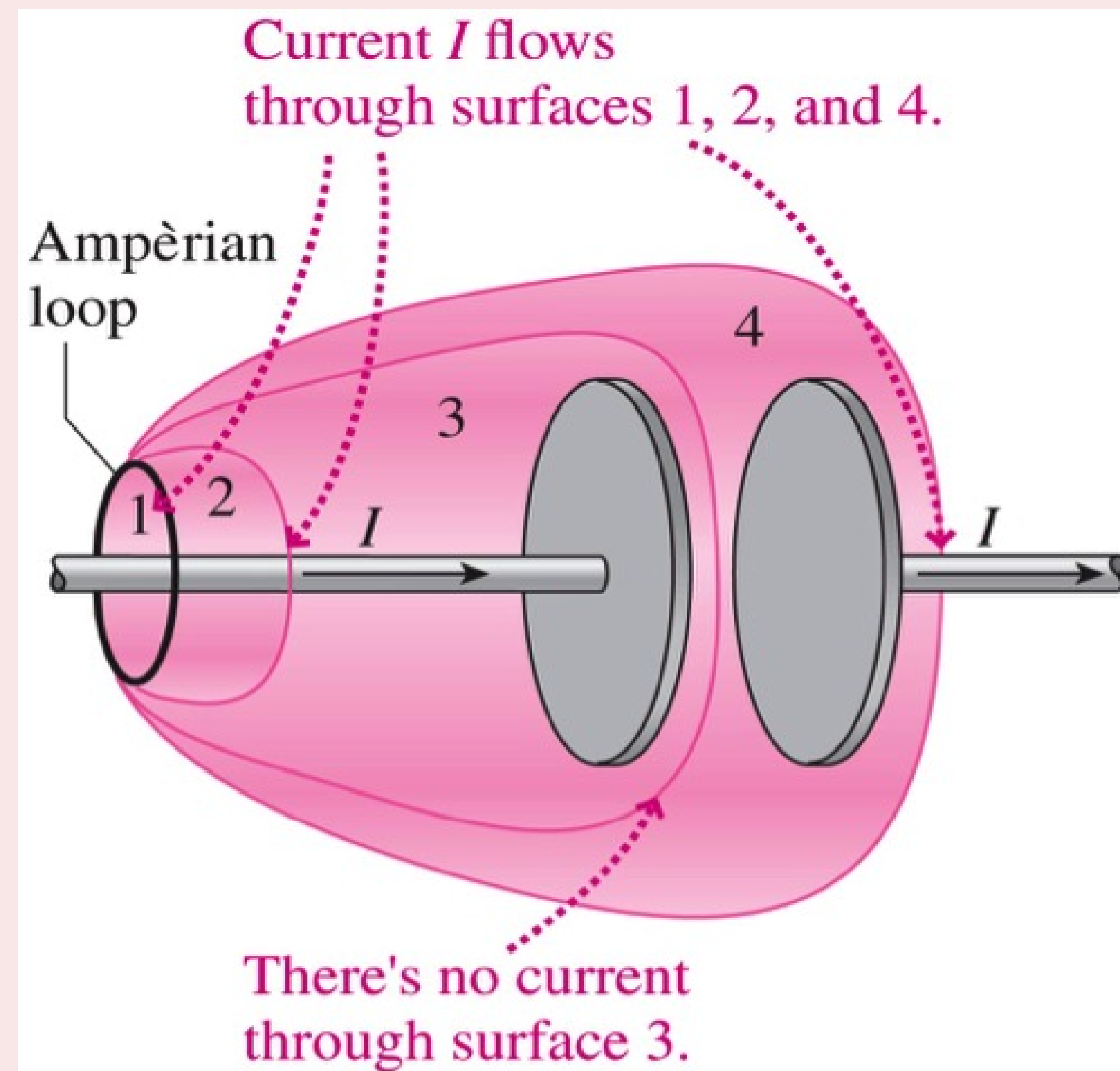


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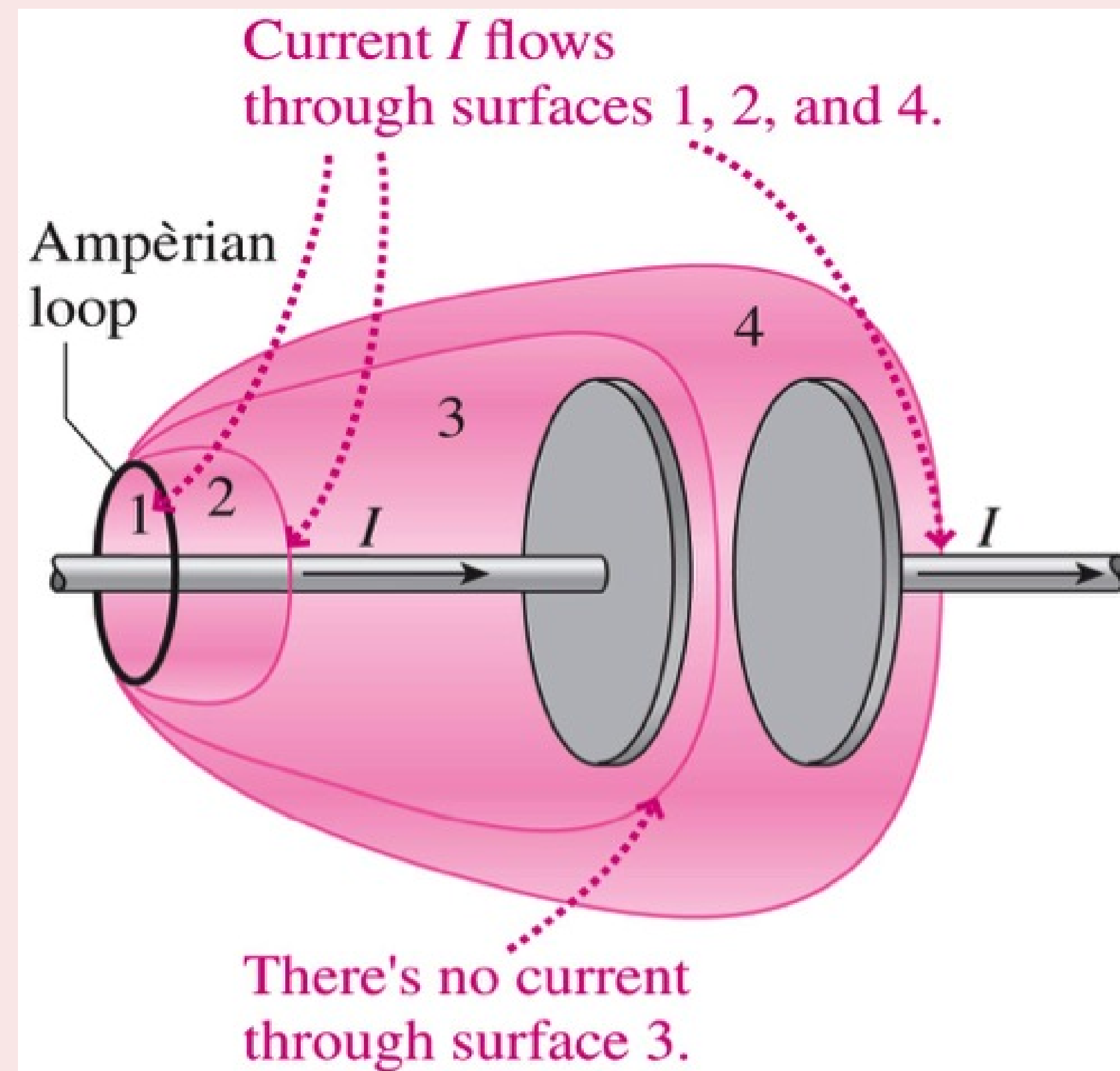


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- The **current given by eq (5) is due to changing electric field (or electric displacement)**. Hence, it is called '**Displacement Current**'.
- The **current carried by conductors due to flow of charges** is called '**conduction current**'.

Ampere-Maxwell law:

- The generalised (and correct) Ampere's circuital law: "the total current passing through any surface of which the closed loop is the perimeter" is the **sum of the conduction current and the displacement current**.

Ampere-Maxwell law:

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- The generalised law is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d) = \mu_0 \left(I_c + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

MAXWELL'S EQUATIONS IN VACUUM

1. $\oint \mathbf{E} \cdot d\mathbf{A} = Q / \epsilon_0$ (Gauss's Law for electricity)
2. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ (Gauss's Law for magnetism)
3. $\oint \mathbf{E} \cdot d\mathbf{l} = \frac{-d\Phi_B}{dt}$ (Faraday's Law)
4. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$ (Ampere – Maxwell Law)

Exercise 1

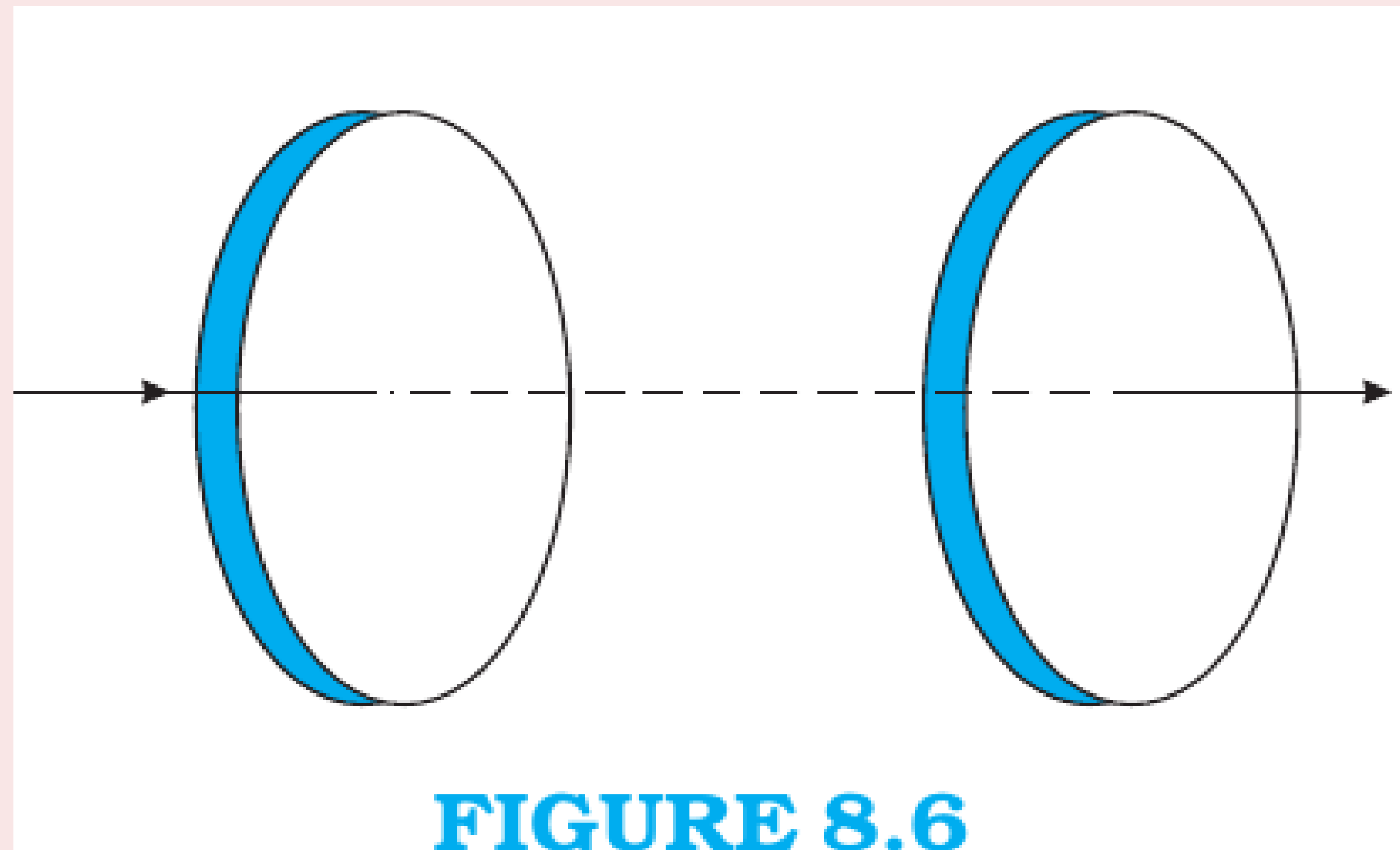


FIGURE 8.6

Figure 8.6 shows a capacitor made of two circular plates each of **radius 12 cm**, and **separated by 5.0 cm**. The capacitor is being charged by an external source (not shown in the figure). The charging **current is constant** and equal to **0.15 A**.

- Calculate the capacitance** and the **rate of change of potential difference** between the plates.
- Obtain the **displacement current** across the plates.
- Is Kirchhoff's first rule (junction rule) valid** at each plate of the capacitor? Explain.

Exercise 2

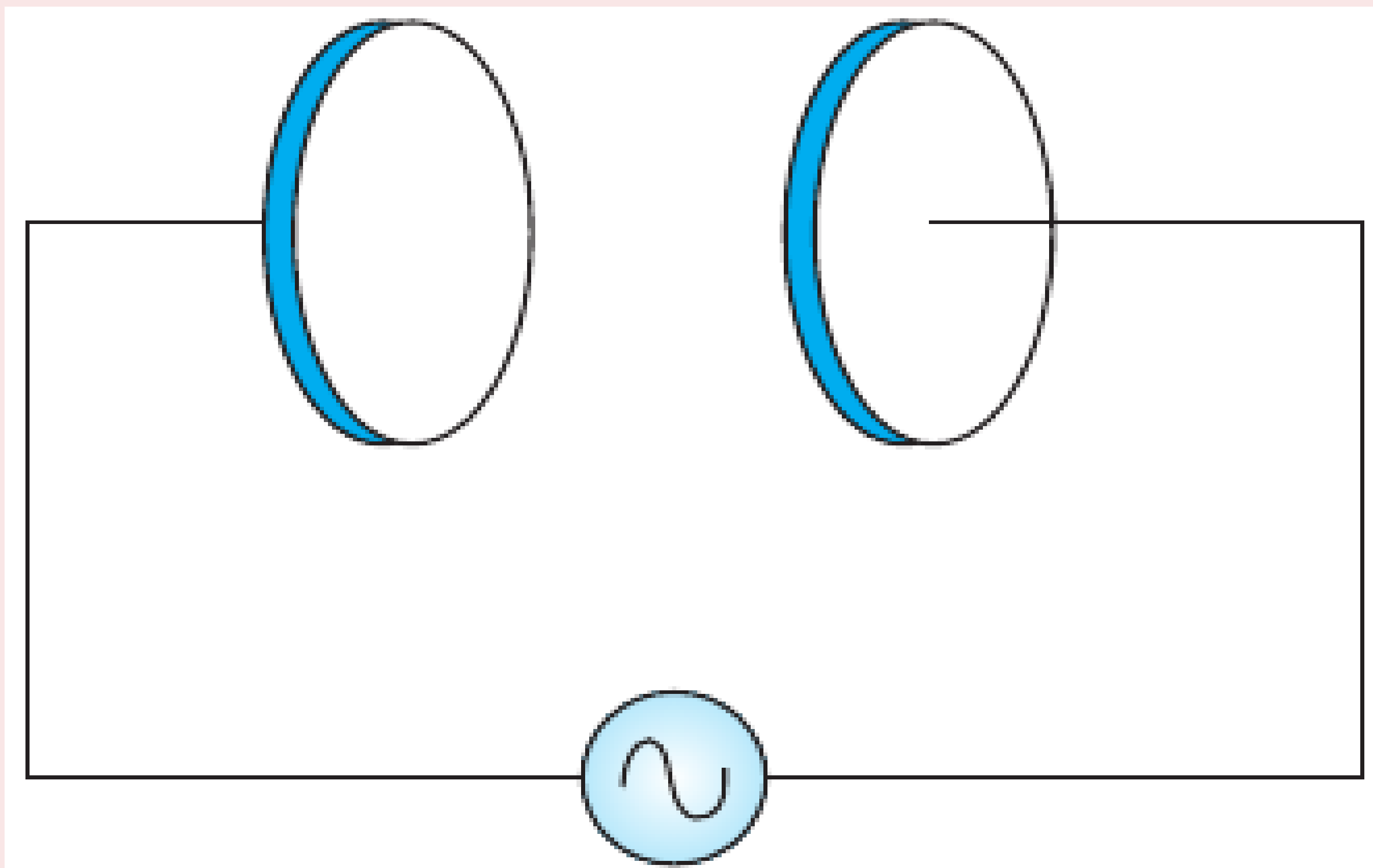


FIGURE 8.7

A parallel plate capacitor (Fig. 8.7) made of circular plates each of **radius $R = 6.0 \text{ cm}$** has a **capacitance $C = 100 \text{ pF}$** . The capacitor is connected to a **230 V ac** supply with a **(angular) frequency** of 300 rad s^{-1} .

- (a) What is the **rms value** of the **conduction current**?
- (b) Is the **conduction current** equal to the **displacement current**?
- (c) Determine the **amplitude of B** at a point **3.0 cm** from the axis between the plates.

ELECTROMAGNETIC WAVES

Sources of electromagnetic waves

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ELECTROMAGNETIC WAVES

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- This **produces an oscillating electric field** in space, which produces an **oscillating magnetic field**, which in turn, is a source of oscillating electric field, and so on.
- The **frequency of the electromagnetic wave** naturally **equals the frequency of oscillation of the charge**.
- The **energy associated with the propagating wave comes** at the expense of the **energy of the source** – the accelerated charge.

ELECTROMAGNETIC WAVES

Sources of electromagnetic waves

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ELECTROMAGNETIC WAVES

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- Seven years after Hertz, **Jagdish Chandra Bose**, working at Calcutta (now Kolkata), succeeded in producing and observing electromagnetic waves of much **shorter wavelength (25 mm to 5 mm)**.

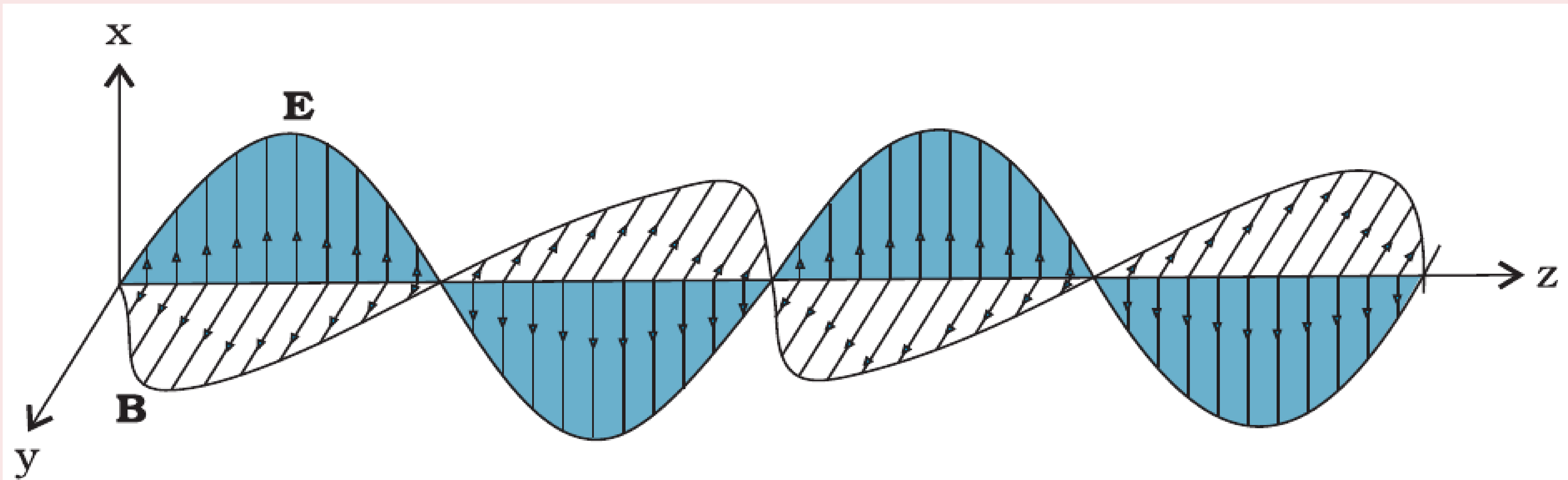
ELECTROMAGNETIC WAVES

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- **Guglielmo Marconi** in Italy followed Hertz's work and **succeeded in transmitting electromagnetic waves** over distances of many **kilometres**.

Nature of electromagnetic waves

- It can be shown from Maxwell's equations that **electric and magnetic fields** in an electromagnetic wave are **perpendicular to each other, and to the direction of propagation**.



Nature of electromagnetic waves

- a typical example of a plane **electromagnetic wave propagating along the z direction**

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- The **electric field $\mathbf{E}$ is along the x-axis**, E_x and varies sinusoidally with z , at a given time.

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- We can write E_x and B_y as follows:

$$E_x = E_0 \sin(kz - \omega t) \quad \text{and} \quad B_y = B_0 \sin(kz - \omega t)$$

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- **speed of wave** $v = \frac{\text{distance}}{\text{time}} = \frac{\lambda}{T} = \lambda\nu$

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- Here, $k = \frac{2\pi}{\lambda}$ is called **propagation or wave vector**.
- $\omega = 2\pi\nu = \frac{2\pi}{T}$ is **angular frequency**
- **speed of wave** $v = \frac{\text{distance}}{\text{time}} = \frac{\lambda}{T} = \lambda\nu$
- Also, $\frac{\omega}{k} = \frac{\frac{2\pi}{T}}{\frac{2\pi}{\lambda}} = \frac{\lambda}{T} = v$

Nature of electromagnetic waves

- Also, speed of EM wave $v = \frac{1}{\sqrt{\mu\epsilon}}$. In Vacuum $c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$

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- EM waves are **Transverse** and are self-sustaining oscillations of electric and magnetic fields in free space, or vacuum. They differ from all the other waves we have studied so far, in respect that **no material medium is involved in the vibrations of the electric and magnetic fields.**

Example 1

A plane electromagnetic wave of **frequency 25 MHz travels** in free space along the **x-direction**. At a particular point in space and time, $E = 6.3 \hat{j}$ V/m. **What is B at this point?**

Example 2

The **magnetic field** in a plane electromagnetic wave is given by
 $B_y = (2 \times 10^{-7}) T \sin (0.5 \times 10^3 x + 1.5 \times 10^{11} t)$.

- (a) What is the **wavelength and frequency** of the wave?
- (b) Write an **expression for the electric field**.

Example 2

Suppose that the electric field amplitude of an electromagnetic wave is $E_0 = 120 \text{ N/C}$ and that its frequency is $\nu = 50.0 \text{ MHz}$.

- (a) Determine, B_0 , ω , k , and λ .
- (b) Find **expressions for E and B**.

Electromagnetic Spectrum

- Classification of EM waves according to frequency/wavelength.

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- They differ in wavelength, frequency, production, detection, and interaction with matter.
- Order of spectrum (increasing frequency): Radio $\rightarrow$ Microwave $\rightarrow$ Infrared $\rightarrow$ Visible $\rightarrow$ Ultraviolet $\rightarrow$ X-ray $\rightarrow$ Gamma.

Radio Waves

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- Detection: receiver antennas.
- Uses: Radio and TV communication, AM/FM, mobile phones (UHF band).

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- Uses: RADAR (navigation, speed guns), microwave ovens (resonant heating of water molecules).

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- Uses: heat therapy lamps, remote controls, weather satellites, greenhouse effect.

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- Note: mostly absorbed by ozone layer; excess exposure harmful.

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- Caution: harmful if overexposed.

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- Detection: same as X-rays.
- Uses: cancer treatment, sterilization.
- Hazard: highly penetrating, dangerous to cells.

Quick Revision Table

Type	Wavelength	Production	Uses
Radio	$> 0.1 \text{ m}$	Accelerated charges in aerials	Radio, TV, mobile communication
Microwave	$0.1 \text{ m} - 1 \text{ mm}$	Klystron, magnetron, Gunn diode	Radar, ovens, satellite communication
Infrared	$1 \text{ mm} - 700 \text{ nm}$	Hot bodies, molecular vibrations	Heat therapy, remotes, weather satellites, greenhouse effect
Visible	$700 - 400 \text{ nm}$	Atomic electronic transitions	Human vision, photographic film, photocells
Ultraviolet	$400 - 1 \text{ nm}$	Sun, special lamps, hot bodies	Sterilization, LASIK surgery, tanning, water purifiers
X-rays	$1 - 10^{-3} \text{ nm}$	Bombard metal target with high-energy electrons	Medical imaging, cancer treatment, metal flaw detection
Gamma	$< 10^{-3} \text{ nm}$	Nuclear reactions, radioactive decay	Cancer therapy, sterilization (high penetration)

Mnemonic for Spectrum

- Easy way to remember order:

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- **“Ravi’s Mother Is Visiting Uncle Xavier’s Garden.”**

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- **“Ravi’s Mother Is Visiting Uncle Xavier’s Garden.”**
- Mapping:
 - Ravi’s → Radio
 - Mother → Microwave
 - Is → Infrared
 - Visiting → Visible
 - Uncle → Ultraviolet
 - Xavier’s → X-rays
 - Garden → Gamma